

Do in-silico metrics filter designed enzymes against catalysis?

Abstract

As designs outnumber what a laboratory can test, in-silico metrics decide which enzymes are built. Pipelines filter on many different metrics, from Rosetta energies and the geometry of the catalytic residues to the confidence of structure-prediction models. Yet no systematic study has asked whether these metrics respond to catalysis or to something simpler, such as proximity to the active site. We therefore designed four tests and applied them to 33 metrics. *Activity discrimination* asks whether a metric separates enzymes that work from enzymes that do not; *substrate discrimination*, whether it knows which substrate an enzyme is built for; *activity ranking*, whether it orders enzymes by their activity; and *detection ability*, whether it notices damage to the catalytic residues more than the same damage elsewhere. Three tests observe natural enzymes and designs as they are, whereas detection ability damages natural enzymes on purpose, so that catalysis can be told apart from proximity to the active site. Every test ranks the metrics beside trivial references, such as the distance to the catalytic residues. In activity discrimination, pLDDT was best on natural enzymes, with an AUROC of 0.79, and the intra-residue repulsion of the catalytic residues on designs, with 0.80. ipTM was best in substrate discrimination, with 0.70, the range of the catalytic pKa in activity ranking, with a Spearman correlation of 0.48, and the per-chain pTM minimum in detection ability, with a probability of 0.75 of changing more at the catalytic residues than at the control.

1 Introduction

Computational enzyme design [1] now produces far more candidates than can be tested. A single design problem yields tens of thousands to millions of designs: 130,000 backbones for a cysteine hydrolase designed with RFdiffusion3 [2], several hundred thousand sequences for a zinc metallohydrolase [3], and 10^5 to 10^6 designs for each starting structure in a Kemp-eliminase workflow [4]. The experiment, one or two 96-well plates, takes up about a hundred, and an in-silico filter decides which. Richter and colleagues described this filtering step in 2011 as "analyzing, and ranking all of the produced structures to find the best handful worth experimentally characterizing" [5].

Recent campaigns have invested heavily in this step. The RFdiffusion3 cysteine hydrolases went through three rounds of filtering on AlphaFold3 [6] predictions of the catalytic geometry, and 18% of the 190 designs tested were multi-turnover enzymes, against 1–2% in earlier campaigns [2]. In the serine-hydrolase campaign, the fraction of designs with ester-hydrolysis activity rose from 1.6% to 5.2% in the round that added a filter for the preorganization of the catalytic residues [7]. In-silico criteria also decide which method is judged better: RFdiffusion2 yields a design that passes them at all 41 active sites of its benchmark, RFdiffusion at 16 [8, 9]. Filtering is thus a decisive step in enzyme design, yet passing it is no guarantee of activity: of the 192 designs tested in the metallohydrolase campaign, 176 have no reported kcat/KM [3].

The filters themselves are far from standardized: what they measure differs from one pipeline to the next. Some rank designs by Rosetta energies and shape complementarity [3, 4, 10], some by how closely a predicted structure reproduces the intended catalytic geometry [2, 11], and others by the confidence of AlphaFold or Chai-1 models of the enzyme–substrate complex [6, 12, 13] or by the preorganization of the active site in an ensemble of predicted conformations [7, 14]. Many pipelines combine several of these into a composite score [2–4]. Cut-offs differ from campaign to campaign, and the weights are set by judgement, in one case as "an arbitrary initial guess" [3].

That judgement has little evidence to draw on: it is not known whether any of these metrics responds to catalysis. A campaign judges its filter by its own hit rate, which also reflects every other change made between rounds. For the final selection, the designers of the metallohydrolases combined many metrics instead of applying cut-offs "due to the lack of known standalone predictive metrics for metallohydrolase activity" [3]. The authors of Riff-Diff, who ranked their designs mainly by the RMSD of the catalytic residues, report a "negligible correlation between predicted enzyme rankings and actual activity in this and previous studies" [11]. Systematic evaluations exist for neighbouring problems: filters for generated enzyme sequences [15], interface

metrics for protein binders [16], predictor confidence for monomer designs [17, 18] and, in 2010, quantum-mechanical and molecular-dynamics scoring of the first Kemp-eliminase designs [19]. To our knowledge, none has asked whether the fast metrics that filter structure-based enzyme designs respond to catalysis.

To address this, we asked what a metric that tracks catalysis should be able to do, and turned each answer into a test. It should tell an enzyme that works from one that does not, which is what a filter is for (*activity discrimination test*). It should also know which substrate an enzyme is built for, as every design conditioned on a substrate assumes (*substrate discrimination test*), and order enzymes by their activity, because a selection ends by choosing the best of the designs that pass (*activity ranking test*). And it should notice damage to the catalytic residues more than the same damage elsewhere in the protein, since that is what separates catalysis from geometry (*detection ability test*). We applied the tests to 33 metrics computable from a design model or a co-folded prediction (Methods 2.1). The first three observe natural enzymes and designs as they are, whereas the fourth damages natural enzymes on purpose, which is what allows catalysis to be told apart from proximity to the active site. Every test ranks the metrics alongside trivial references, such as the distance to the catalytic residues, so that a metric is credited only if it does better than something simple.

In activity discrimination on natural enzymes, pLDDT ranked first (AUROC 0.79) and the best of the 33 metrics, the intra-residue repulsion of the catalytic residues, reached 0.61, against a chance threshold of 0.68 for the best of 39 items. On the designs the intra-residue repulsion ranked first (0.80) and the tyrosine fraction reached 0.75. In substrate discrimination, ipTM ranked first (0.70) and the RMSD of the catalytic atoms reached 0.53 to 0.56. In activity ranking, the range of the catalytic pKa ranked the BglB variants first (Spearman $\rho = 0.48$), the distance from the mutated residue to the catalytic residues reached 0.46, and no metric beat it. In detection ability, the per-chain pTM minimum had the highest R (0.75; 0.5 means no preference for the catalytic site) and an R of 0.59 when the control lay at the same distance from the ligand (Results 3.1 to 3.4).

2 Methods

2.1 Experiment overview

Four tests, one in each of 2.2 to 2.5, ask whether the in-silico metrics used to filter designed enzymes respond to catalysis or to something simpler, such as how close a change is to the active site:

- **Activity discrimination test (2.2; results in 3.1).** The metrics are ranked by how well they separate real dead from active proteins, 21 zymogen–mature enzyme pairs and 192 plated de novo designs of which 16 are active.
- **Substrate discrimination test (2.3; results in 3.2).** A structure predictor is given each of 55 natural enzymes with its own substrate and with a wrong ligand, and the prediction-based metrics are ranked by how well they separate the two predictions.
- **Activity ranking test (2.4; results in 3.3).** The metrics are ranked by how well their order of 432 BglB variants and of 16 plated designs agrees with the measured activity, against declared trivial baselines.
- **Detection ability test (2.5; results in 3.4).** Catalytic residues are deliberately damaged in up to 195 natural enzymes, and the metrics are ranked by whether they move more than when the same damage is made at a matched non-catalytic site of the same protein.

Metrics. The same panel is scored in every test: 29 structure-space metrics, which read atomic coordinates and are used in 2.2, 2.4 and 2.5, and 4 prediction-based metrics, which read the output of a structure predictor that is given a sequence and a ligand and are used in 2.2 to 2.5 (Table 1). Numbers in square brackets are references (see References); tables with an S number are supplementary tables, described in the Supplementary PDF and released as CSV files (main/ and supplementary/).

Table 1. The 33 reported metrics and the six reference rows. In a structure, "perturbed residues" are the residues that were changed; in the control arm they are the control residues, so a site-scoped metric never knows which residues are catalytic — it only knows which residues it was handed. In 2.2 and 2.4 the scored residues are the catalytic residues. The third column says where each definition comes from.

Group	Metrics	Reads, and where it is defined	Scored over
Rosetta energy, site-scoped (15)	attraction, repulsion, intra-residue repulsion, solvation, lk-ball solvation, electrostatics, hydrogen bonds (backbone–side-chain, side-chain–side-chain), rotamer (Dunbrack), Ramachandran, amino-acid propensity, omega torsion, net van der Waals (repulsion plus attraction), site total, worst single residue	coordinates; the per-residue terms of the REF2015 energy function [20] computed with PyRosetta [21] and summed over the scored residues (site total: the sum of the residue totals; worst residue: the largest residue total)	the perturbed residues

Group	Metrics	Reads, and where it is defined	Scored over
Catalytic geometry (8)	fraction buried, relative solvent-accessible area (mean, max), total solvent-accessible area, pairwise distance (mean, max), radius of gyration, polar contacts	coordinates; SASA by the Lee–Richards algorithm [22] in FreeSASA [23] (probe 1.4 Å, waters excluded), relative to the maximum accessibility of the residue type [24]; fraction buried is the share of the scored residues with relative SASA below 0.10; distances and radius of gyration are over all pairs, or all atoms, of the side-chain heavy atoms of the scored residues (backbone atoms for Gly); a polar contact is a pair of polar side-chain atoms of two scored residues within 3.5 Å	the perturbed residues
Catalytic pKa (6)	PROPKA pKa (min, max, mean, range) and pKa shift (mean, max absolute)	coordinates; PROPKA3 [25, 26] pKa of the scored ionisable residues; the shift is the pKa minus the pKa of the model compound	the perturbed residues
Interface confidence (2)	ipTM; per-chain pTM minimum	predictor output of Chai-1 [13]; ipTM as introduced for AlphaFold-Multimer [27]; the pTM of each chain, from the TM-score head of AlphaFold [12]	the protein–ligand interface
Active-site accuracy (2)	AME RMSD (catalytic-atom RMSD to a reference structure after backbone superposition, best of the predicted models); ligand clearance (closest approach of the ligand to the backbone, best of the models)	predictor output and a reference structure; the criteria of the Atomic Motif Enzyme benchmark of RFDiffusion2 [8], as implemented in this project (equations in 2.3)	the catalytic atoms; the ligand
Reference rows (6; never counted as metrics under test)	resolution, sequence length, net charge, tyrosine fraction, pLDDT, pTM	deposit, sequence or predictor; net charge is $(K + R) - (D + E)$ counted in the sequence; pLDDT and pTM from AlphaFold [12]	whole protein

Which metrics are reported. The panel contains 216 named quantities, which collapse to 191 distinct metrics once literal aliases are merged; 63 of them are bookkeeping counters and constants on which no test can be run. Whether a metric can be tested in an experiment is decided from what the metric reads, whether it is defined under the experimental condition and the design of the experiment, never from the result. The main text reports the 33 metrics that are valid and tested in all four tests (Table 1), so that one set of metrics runs through the whole paper, plus six reference rows chosen before any result was seen. The Chai-1 combined score, a rescaled ipTM and not an independent metric, is scored but kept out of the main set. The role of each of the 191 distinct metrics and the reason for it are in Table S1; what each of the 216 named quantities reads, and over which residues it is scored, is in Table S2; whether each can be tested in each of the 30 tests, and if not why, is in Tables S3 (one row per metric and test) and S4 (counts by class of metric); the counts of the earlier whole-panel analysis restated on the eligible metrics only are in Table S5; and the list of the tests, with comparator, control type and role, is in Table S6.

Notation. Every Results table ranks a set of *items* (the metrics, the reference rows and, in 2.4, the declared baselines): m is an item, M the number of items ranked in one table and $x_u(m)$ the value of item m on a unit u (a structure, a prediction, a variant or a design). Bootstrap and permutation tests use $N_{\text{boot}} = N_{\text{perm}} = 5,000$ draws unless another number is stated; q_α is the α -quantile of a set of values and $\mathbf{1}(\cdot)$ is 1 if its argument is true and 0 otherwise. The statistics (A_m , ρ_m , $R_{m,s}$ and the best-of- M threshold from the max-statistic $T^{(r)}$) are defined with their equations in 2.2 to 2.5.

2.2 Activity discrimination test: do the metrics separate a dead enzyme from an active one?

Experiment. Every metric is scored on real proteins in which one state is catalytically dead and the other active, and the metrics are ranked by how well their values separate the two states. A zymogen, for example, is an inactive precursor that already contains the catalytic residues of the mature enzyme: a metric that detected catalytic competence would separate the two structures, whereas a metric that only reads the geometry of the catalytic residues would not, because that geometry is almost the same in both. Two datasets are used, each with its own table and its own n : 21 zymogen–mature pairs (2.2.1; results in 3.1.1, Table 5) and 192 plated designs, of which 16 are active (2.2.2; results in 3.1.2, Table 6). The procedure is the same for both:

1. assemble the structures and label each one (dead or active; active or no active);
2. compute every item on every structure: the structure-space metrics on the deposited coordinates, the prediction-based metrics on a Chai-1 [13] prediction of the structure together with its ligand;
3. compute the AUROC of every item and a bootstrap interval that resamples the clustering unit;
4. rank the items by the direction-free AUROC and compare the top of the ranking with the best-of- M permutation threshold.

2.2.1 Discriminate natural active and dead enzymes

Dataset. Pairs of structures of the same protein, the zymogen (dead) and the mature enzyme (active). The structures were downloaded from the RCSB Protein Data Bank [28, 29]; the proteins and the propeptide boundaries come from UniProt [30]. The pairs were assembled in the steps of Table 2, and the evaluation set is the last:

Table 2. Assembly of the zymogen–mature pairs. From the UniProt search to the 21-pair evaluation set of Results 3.1.1; the last row is the evaluation set.

Step	Cases	Rule
UniProt discovery	reviewed proteins of EC 3.4.* with a propeptide feature and PDB structures	a protein is shortlisted when its PDB chain ranges give both a candidate zymogen and a candidate mature X-ray entry
Shortlisted proteins	100	the proteins that pass the rule above
PDB entities classified	5,717 (1,182 zymogen, 4,528 mature, 7 indeterminate)	each entity is called zymogen or mature from its SIFTS [31] alignment to the UniProt chain and propeptide boundaries, never from the entry title
Candidate pairs	55 pairs of 55 proteins	one zymogen and one mature entity per protein; both X-ray, no engineered mutation, resolution at most 2.5 Å
Scored pairs (wider set)	49	structure-space metrics scored on both forms (S7); trapping-mutant pairs were examined and retired
Re-predicted pairs	28	both forms predicted with Chai-1, three seeds each (S8)
Evaluation set	21	the two forms share a ligand; the 7 apo pairs are excluded

The 21 pairs are 21 proteins, all peptidases (9 serine, 8 cysteine and 4 aspartic; EC 3.4.21, 3.4.22, 3.4.23), with shared ligands such as inhibitors and metal ions. In the wider set of 49 pairs the median catalytic-constellation RMSD between the two forms is 0.207 Å (0.484 Å after backbone superposition). The definitions of the six pair sets used in 3.1 (55, 49, 45, 28, 21 and 96 pairs) and the table that uses each are in Table S9.

Why this dataset. A zymogen and its mature enzyme hold the catalytic residues constant while the enzyme is dead in one form and active in the other, so they are a *real* negative: the labels are correct by definition. They answer the objection that a metric may only recognise synthetic negatives. The 21-pair subset is the evaluation set because it is the only one on which every main-set metric, structure-space and prediction-based, and every reference row has a value on both forms of every pair, so that one n applies to every row of Table 5. The pairs were selected by ligand sharing and not by any metric value; the choice was made after the 49-pair and 28-pair analyses had been read, and those analyses are in the Supplement (S7, S8). A second class of dead enzymes, substrate-trapping mutants, was verified against the primary literature (26 of 30 were not confirmed dead) and retired (S10).

Metrics scored. The 33 main-set metrics (Table 1) and the 6 reference rows, 39 items in all, are computed on both forms of each pair. The 29 structure-space metrics are computed on the deposited coordinates over the catalytic residues, taken from M-CSA [32] and UniProt [30] annotation. For the 4 prediction-based metrics both forms of each pair were re-predicted with Chai-1 together with the shared ligand (three seeds, averaged), and active-site accuracy is measured against each form's own deposited structure. Table S11 lists every metric scored on the 21 pairs (158, including whole-protein, sequence-only and PLACER [14] comparators). The reference rows are reported in full, with intervals, for every activity test in Table S12.

Ranking criteria. Let P be the dead forms and Q the active forms, $|P| = |Q| = n = 21$. The AUROC of item m is the probability that a randomly chosen dead form has a larger value than a randomly chosen active form, ties counting one half [33]:

$$A_m = \frac{1}{|P||Q|} \sum_{p \in P} \sum_{q \in Q} \left[\mathbf{1}(x_p(m) > x_q(m)) + \frac{1}{2} \mathbf{1}(x_p(m) = x_q(m)) \right] \quad (1)$$

This equals the Mann–Whitney statistic U divided by $|P||Q|$ [34]. $A_m = 0.5$ means that the item does not separate the two forms, $A_m = 1$ that every dead form is above every active form and $A_m = 0$ that every dead form is below every active form. We ask whether an item separates the forms, not in which direction, so items are ranked by

$$A_m^* = \max(A_m, 1 - A_m) \geq 0.5, \quad (2)$$

and the directed A_m is kept so that the direction is visible ("higher in" in Table 5 is the form in which the item takes larger values: dead if $A_m > 0.5$). Because A_m^* cannot fall below 0.5, the lower end of its interval is close to 0.5 for almost every item and is not informative by itself.

The interval is a percentile bootstrap that resamples **pairs** with replacement, so that both forms of a drawn pair enter together [35, 36]. With $b = 1, \dots, N_{\text{boot}}$ draws (5,000, seed 20260806), $A_m^{*(b)}$ is recomputed on each resampled set and

$$\left[q_{0.025}(A_m^{*(1)}, \dots, A_m^{*(N_{\text{boot}})}), q_{0.975}(A_m^{*(1)}, \dots, A_m^{*(N_{\text{boot}})}) \right] \quad (3)$$

is the 95% interval. A pair is also a protein, so the design-level and the cluster-level N are both 21.

Table 5 ranks $M = 39$ items, and the top of such a ranking looks good by chance. To say how good, the labels are permuted. Within a pair the two forms are exchangeable if the item is unrelated to the label, so in replicate $r = 1, \dots, N_{\text{perm}}$ we draw $s_i^{(r)} \sim \text{Bernoulli}(1/2)$ for every pair i , swap the dead and active labels of the pair when $s_i^{(r)} = 1$, and recompute $A_m^{*(r)}$ for every item. The same swaps serve all items, which keeps the correlation between metrics. The largest value over the items in each replicate,

$$T^{(r)} = \max_{m=1, \dots, M} A_m^{*(r)}, \quad (4)$$

has the distribution of the best AUROC of M unrelated items. Its median and its 95th percentile $q_{0.95}(T)$ are the **best-of-39 threshold** (median 0.63, 95th percentile 0.68): an observed A_m^* above it is higher than the best of 39 unrelated items in 95% of simulations. Taking the maximum over items is the max-statistic form of permutation control of multiple comparisons [37]. The threshold is a reference level for reading the ranking. For the complete table of all 158 metrics (S11) the same procedure gives the best-of-158 threshold. In Table 5 the AUROCs above the threshold are in bold. No threshold is adopted for this set: we report whether any main-set metric exceeds the best-of-39 threshold, and no point estimate is offered as a headline when its interval is wider than 0.2.

2.2.2 Discriminate de novo designed active and no active enzymes

Dataset. A deposited metallohydrolase design campaign [3]: 192 designs on two 96-well plates (two campaigns of 96 designs, the two 96-design zinc campaigns that are also reported with RFDiffusion2 [8]). Of the 192, **16 have a reported kcat/KM (active)** and **176 are no active**. The designs lie in **136 backbone clusters**, and the 16 active designs lie in 8 of them (5 in each of two clusters and one in each of six others). The catalytic motif is the three histidines that coordinate the zinc, derived geometrically and checked against 23 curated annotations (23 of 23 agree). For the predictions the ligand is the design's transition-state analogue together with the zinc. One set serves every row of Table 6: every item is scored on all 192 designs.

Why this dataset. These are the deposited de novo enzyme designs for which models and measured activity are both available, so they let the metrics be tested on designs and not only on natural enzymes. A design absent from the table of characterised hits was not reported as active, and the comparison is active against no active. The plate analyses were planned as descriptive, without an AUROC, before the data were analysed; the descriptive analyses (rank correlation among the 16, hits per plate) are reported in 2.4.2 and the AUROC, added later, in this subsection.

Metrics scored. The 36 items are the 29 structure-space main-set metrics, the two prediction-based main-set metrics that the plates define (ipTM and the per-chain pTM minimum) and five reference rows (tyrosine fraction, net charge, sequence length, pLDDT and pTM; designs have no crystallographic resolution). An item must have a value on at least 80% of the designs. The AME RMSD and the ligand clearance are measured on the plates against each design's own model, which is a different metric from AME against a deposited structure (the two flavours are never pooled), so they are in the Supplement and not among the ranked items. Table S13 lists every metric scored on the designs (128).

Ranking criteria. The statistic is Eq. (1) and (2) with P the 16 active designs and Q the 176 no active designs, over the 192 designs; "higher in" in Table 6 is the group in which the item takes larger values. The interval resamples the $G = 136$ **backbone clusters** with replacement, all designs of a drawn cluster entering together, so that designs built on one backbone are not counted as independent (5,000 draws, seed 20260807; a draw with no active design has no AUROC and is dropped):

$$\left[q_{0.025}(A_m^{*(b)}), q_{0.975}(A_m^{*(b)}) \right], \quad A_m^{*(b)} = \max(A_m^{(b)}, 1 - A_m^{(b)}), \quad (5)$$

with $A_m^{(b)}$ from Eq. (1) on the designs of the drawn clusters. The design-level N is 192 and the cluster-level N is 136, with 16 actives in 8 clusters. The chance reference is the permutation of Eq. (4) with the activity labels permuted among the 192 designs instead of swapped within pairs, $M = 36$ items and 5,000 permutations: the **best-of-36 threshold** (median 0.66, 95th percentile

0.73), and the best-of-128 threshold (0.75) for the 128 metrics of Table S13. Because 10 of the 16 active designs lie in two backbone clusters, designs are not exchangeable and the threshold understates the chance range. Table 6 marks the AUROCs above the best-of-36 threshold in bold.

2.3 Substrate discrimination test: do prediction-based metrics know which ligand they were given?

Experiment. A structure predictor (Chai-1 [13]) is given a protein together with its own substrate, and then together with a wrong one, and the prediction-based quantities are ranked by how well they separate the cognate prediction from a wrong-ligand prediction of the same enzyme. For example, Chai-1 predicts a hydrolase together with its own substrate and again together with a substrate of a different enzyme class; if ipTM falls when the wrong substrate is given, the metric "knows" which substrate belongs to the protein, and the AUROC measures how reliably it does. Chai-1 is run on the sequence, the ligand (as a chemical structure) and, where relevant, the catalytic metal, with default settings and a fixed seed; each prediction yields five models and their confidence scores. Only the ligand supplied changes; the protein sequence is the same in every condition, and four conditions are built for every enzyme (results in 3.2, Table 7):

1. **cognate:** the substrate of the enzyme (three seeds, 101, 202 and 303; the cognate condition has two further seeds that serve as the noise floor and are not used in the evaluation);
2. **same class:** a different substrate of an enzyme of the same enzyme class;
3. **different class:** a substrate of an enzyme of a different class;
4. **decoy:** a ligand taken from an unrelated structure.

Wrong ligands have three seeds each, and the catalytic metal is kept in every condition. The apo condition (no ligand) is excluded for interface and ligand-dependent terms, because without a ligand there is no interface to be confident about.

Dataset. Natural enzymes of M-CSA [32] with a bound cognate ligand: 59 enzymes enter the experiment, and the **evaluation set is the 55 natural enzymes** on which every scored quantity has a defined value in the cognate condition and in all three wrong-ligand conditions, and on which every ligand that was supplied was present in the prediction. The four exclusions are enzymes whose cognate or wrong ligand contains a heme-type group that cannot be built into a chemical structure, so the predictor drops it and the prediction is effectively apo for that cell. The rule is about the availability of values and not about any result. The experiment is balanced: seeds 101, 202 and 303 in every condition. Twenty-eight of the 59 enzymes are also in the 143-enzyme set of 2.5.

Why this dataset. A wrong ligand can only be defined for an enzyme whose substrate is known, and the M-CSA entries with the cognate reaction ligand present (150 of 434) are the natural enzymes for which this holds. Natural enzymes carry the main analysis because their active-site accuracy is measured against the deposited structure; the 30 de novo designs, whose active-site accuracy is measured against the design's own model and is never pooled with it, are in S14. The choice of the 55 systems was made after the earlier all-system analysis had been read.

Metrics scored. Every prediction is scored on 33 prediction-based quantities, of which Table 7 ranks 30. The three summaries of the Chai-1 combined score are scored but not listed, because the combined score is not a main-set metric (see "Which metrics are reported"); their AUROCs are in S14. The 30 listed quantities are the four main-set metrics of Table 1, 13 whole-prediction confidence summaries (pLDDT, pTM, the per-chain pTM of the protein chain and the inter-chain clash flag, two of which, the mean pLDDT and mean pTM, are the reference rows), the AME pass bit, and four bookkeeping counters (models, aligned and catalytic atoms, symmetry swaps) that serve as negative controls. Each score s_k is returned for each of the $K = 5$ models, and every score is summarised over the models as

$$\bar{s} = \frac{1}{K} \sum_{k=1}^K s_k, \quad s_{\max} = \max_k s_k, \quad s_{\text{SD}} = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (s_k - \bar{s})^2}, \quad (6)$$

and the RMSD also as the minimum over the models, so that one metric appears in several rows of Table 7, one per summary. Two quantities need a definition of their own (the other summaries of whole-prediction confidence are those of Chai-1 [13]).

AME RMSD (active-site accuracy; the criterion of the Atomic Motif Enzyme benchmark [8]). A model is superposed on the reference structure by the Kabsch algorithm on the $C\alpha$ atoms of all residues shared with the reference (the whole shared backbone, and not the catalytic residues alone, which would let the site translate freely inside the fold and deflate the RMSD). The RMSD of the catalytic atoms is then

$$\text{RMSD}_k = \sqrt{\frac{1}{L} \sum_{l=1}^L \|\mathbf{y}_l - \mathbf{x}_l^{(k)}\|^2}, \quad \text{AME RMSD} = \min_k \text{RMSD}_k, \quad (7)$$

where $l = 1, \dots, L$ runs over the heavy atoms of the catalytic residues that keep their residue type in the prediction, $\mathbf{y}_l$ is the atom in the reference, $\mathbf{x}_l^{(k)}$ the superposed atom of model k , and symmetry-equivalent atoms (the two oxygens of a carboxylate, for example) are assigned to give the smaller RMSD. The reference is the deposited structure for natural enzymes and the design's own model for designs; the two flavours are never pooled. The variant without symmetry resolution is also listed ("no symmetry resolution" in Table 7).

Ligand clearance is the closest approach of the ligand to the protein backbone atoms N, C α and C, best case over the models:

$$C_k = \min_{a \in \text{ligand}} \min_{b \in \{\text{N}, \text{C}\alpha, \text{C}\}} \|\mathbf{x}_a^{(k)} - \mathbf{x}_b^{(k)}\|, \quad \text{clearance} = \max_k C_k, \quad (8)$$

so that a larger value means more room for the ligand (the AME criterion is $C > 1.5$ Å). Interface confidence (ipTM, per-chain pTM minimum) and ligand clearance need a ligand and are undefined without one.

Ranking criteria. For each quantity m and each kind of wrong ligand $c \in \{\text{same class, different class, decoy}\}$, the AUROC of Eq. (1) is computed with the cognate predictions as the positive class and the wrong-ligand predictions of kind c as the negative class, one value per system (the mean over its three seeds), and made direction-free as in Eq. (2); quantities are ranked by the mean of the three,

$$\bar{A}_m = \frac{1}{3} \sum_c A_{m,c}^*. \quad (9)$$

Each $A_{m,c}^*$ has the percentile bootstrap interval of Eq. (3) that resamples systems (5,000 draws, seed 20260806); "higher in" is the condition in which the quantity takes larger values. The chance reference is the max-statistic permutation of Eq. (4): in each of 5,000 replicates the four condition labels (cognate, same class, different class, decoy) are permuted at random within every system, the three AUROCs and their mean are recomputed for every quantity, and $T^{(r)} = \max_m \bar{A}_m^{(r)}$ over the $M = 30$ ranked quantities; its median and 95th percentile are the **best-of-30 threshold** (median 0.56, 95th percentile 0.61) [37]. Table 7 marks in bold the mean AUROCs above it. No threshold is adopted; we report which quantities exceed it.

To check whether a ligand-size effect explains the discrimination, the cells (a cognate ligand and one wrong ligand of an enzyme) are stratified by the difference in heavy-atom count, $\Delta h = h_{\text{cognate}} - h_{\text{wrong}}$, and in charge, and the paired win fraction of the quantity is reported (S15):

$$W = \frac{1}{n_{\text{cells}}} \sum_{\text{cells}} \left[\mathbf{1}(x_{\text{cognate}} > x_{\text{wrong}}) + \frac{1}{2} \mathbf{1}(x_{\text{cognate}} = x_{\text{wrong}}) \right]. \quad (10)$$

2.4 Activity ranking test: do the metrics track measured activity?

Experiment. We ask whether the order of real variants (or real designs) by a metric agrees with their order by measured activity, and whether the metric does better than simply knowing how close the change is to the active site. For example, the variants are lined up by how strongly the assay found each one impaired and again by how much a metric changes for each one; if the two orders agree, the metric ranks the variants as the assay does, and the rank correlation measures how well. The baselines ask whether knowing only the distance from the mutated residue to the catalytic glutamates ranks them as well. This is a ranking claim, so the statistic is a rank correlation; the AUROC of the binary contrast (impaired against wild-type-like variants) is given for BglB in the Supplement (S16). Two datasets are used, each with its own table and its own n: 432 BglB variants (2.4.1; results in 3.3.1, Table 8) and the 16 plated designs that have a measured activity (2.4.2; results in 3.3.2, Table 9).

2.4.1 Ranking natural enzyme activity

Dataset. β -glucosidase B (BglB) from *Paenibacillus polymyxa*, characterised by the D2DCure consortium [38] (data retrieved from d2dcure.com on 2026-08-06; the licence is unstated, so variant-level data are not redistributed). Variant numbering is offset by three from the numbering of the crystal structure (RCSB PDB entry 2JIE [28, 29]; structure position = variant position + 3).

The catalytic residues are Glu167 (proton donor) and Glu356 (nucleophile) in structure numbering, taken from the UniProt [30] active-site features, the conserved motifs TINEP and ITENG, and the covalent link to the glucosyl intermediate in the deposited structure. The funnel from the measured rows to the evaluation set is in Table 3:

Table 3. From the D2DCure rows to the 432-variant evaluation set. The funnel for Results 3.3.1; the last row is the set on which every row of Table 8 is computed.

Step	Cases	Rule
Measured rows	1,288 rows, of which 272 are wild-type replicates measured at eight institutions	the D2DCure file (latin-1 encoded)
Single-point variants	655 at 207 positions	the variants of the characterisation data
Set aside	108 destabilised, 83 non-expressed, 32 with insufficient data	a variant is folded if it was expressed and is not destabilised (T50 or Tm less than 2 standard deviations below wild type); the kinetics of the others cannot be attributed to the catalytic site
Folded variants with kinetics	432 at 175 positions: 166 impaired, 254 wild-type-like, 12 improved	impaired if kcat/KM is at least 2 standard deviations (a 3.8-fold loss or more) below wild type, wild-type-like if within 2 standard deviations, improved above
Evaluation set	432 variants	one set serves every row of Table 8; a structure-space metric has a value for 431 of the 432 variants and a prediction-based metric for all 432; the dataset’s own Rosetta score exists for 248

The spread of the wild-type replicates (0.293 log10) is the noise floor, and each variant is compared with the wild type of the same institution. The binary contrast of the Supplement (S16) uses the 420 variants that are impaired or wild-type-like. Substrate: the prediction-based metrics use the assay substrate 4-nitrophenyl β -D-glucopyranoside (pNPG), the reporter substrate of the published BglB assays [38, 39], and no metal, because BglB is metal-independent.

Why this dataset. BglB is the powered calibration set of natural-enzyme variants: hundreds of single-point variants of one enzyme with kinetics, expression and thermal stability (T50, Tm) measured in one consortium, so that a ranking claim can be tested at a scale that no set of designs offers, and so that a loss of activity from a failure to fold (a destabilised or non-expressed variant) can be separated from a loss at the catalytic site. The analysis plan was committed and dated before any mutant structure was built and before any metric was joined to a label.

Metrics scored. Each variant is built in silico with the same rotamer-space substitution, repack and minimisation as in 2.5 (the wild type goes through the identical treatment with five seeds), and the metrics are computed over the two catalytic glutamates for every variant. The prediction-based metrics are computed on Chai-1 predictions of each variant and of the wild type with the assay substrate; AME is measured against the wild-type crystal, which carries a covalent glucosyl intermediate and not the substrate. The 41 ranked items of Table 8 are the 28 structure-space main-set metrics that BglB defines (the omega term does not change under a side-chain substitution on a fixed backbone), the 4 prediction-based main-set metrics, four reference rows (tyrosine fraction, net charge, pLDDT and pTM; the sequence length is constant over single substitutions and cannot be ranked) and five **declared baselines**, fixed before any metric was joined to a label, each pointed so that a larger value means more impaired: the absolute change in side-chain volume; the negative distance from the mutated residue to the nearest catalytic residue in the wild-type structure (the *distance baseline*); the negative relative solvent-accessible area of the mutated residue (burial); the negative BLOSUM62 score of the substitution [40]; and the Rosetta score that comes with the dataset (available for 248 of the variants, so compared on that subset).

Ranking criteria. The measured **impairment** of variant v is the log-ratio of its kcat/KM to the wild-type value of the same institution,

$$I_v = \log_{10} \frac{\kappa_{wt}}{\kappa_v}, \quad (11)$$

larger for a more impaired variant (κ is kcat/KM). A metric enters as the size of its change, the absolute difference between the variant and the treated unsubstituted protein, $|\Delta_v(m)| = |m(v) - m(wt)|$, and a declared baseline enters as its own score. The direction is fixed in advance (a larger change in a more impaired variant), so the directed Spearman correlation is reported and a negative value is kept as it is. With r_v the rank of the item’s score and s_v the rank of I_v among the $n = 432$ variants (ties take the average rank), the Spearman rank correlation is the Pearson correlation of the ranks [41]:

$$\rho_m = \frac{\sum_v (r_v - \bar{r})(s_v - \bar{s})}{\sqrt{\sum_v (r_v - \bar{r})^2 \sum_v (s_v - \bar{s})^2}} \quad (12)$$

$\rho = 1$ means that the item orders the variants exactly as the assay does, $\rho = 0$ that the two orders are unrelated, and $\rho = -1$ that they are opposite. The 95% interval is a percentile bootstrap (Eq. 3 with ρ in place of A^*) that resamples the 175 **positions** with replacement (5,000 draws, seed 20260807), because variants at one position are not independent; the design-level N is 432 and the cluster-level N is 175. To say whether an item is better than the distance baseline, the difference is computed on the same draws,

$$D_m^{(b)} = \rho_m^{(b)} - \rho_{\text{dist}}^{(b)}, \quad (13)$$

and the item **beats** the baseline only if the 95% interval of D_m , $q_{0.025}(D_m^{(b)})$ to $q_{0.975}(D_m^{(b)})$, lies above 0. Table 8 ranks $M = 41$ items by ρ_m , and the top of such a ranking looks good by chance, so the measured impairments are permuted among whole positions that have the same number of variants (a block permutation, so that the variants at one position keep their impairments together, as in the bootstrap), $\rho_m^{(r)}$ is recomputed for every item with the same permutation, and

$$T^{(r)} = \max_{m=1, \dots, M} \rho_m^{(r)}, \quad r = 1, \dots, N_{\text{perm}}. \quad (14)$$

The median and the 95th percentile of T are the **best-of-41 threshold**: the correlation that the best of 41 unrelated items would exceed in 5% of permutations (median 0.24, 95th percentile 0.32; for the complete table S16, over all 114 analysed quantities, 0.32). The same procedure applies to the declared baselines, whose raw score takes the place of $|\Delta|$. It is a reference level for reading the ranking; a ρ above it is in bold in Table 8. The signed correlation of the pre-specified plan (S17) and the AUROC of the binary contrast (S16) are given beside it. The ten sensitivity analyses (variants near to and distal from the catalytic residues, restricted subsets of variants or metrics, other thresholds for the labels, and each institution left out in turn) are summarised in Table S18, the 28 prediction-based metrics, with their own correlation and AUROC columns, are in Table S19, and the declared baselines are reported in full in Table S12.

2.4.2 Ranking de novo designed enzyme activity

Dataset. The 192-design campaign of 2.2.2 [3]. Only the 16 designs with a reported kcat/KM have a measured activity that an item can be correlated with, so the **evaluation set is those 16 designs** and one n serves every row of Table 9. They lie in 8 of the 136 backbone clusters, so the effective number of independent designs is smaller than 16. The 192-design plate is used for the AUROC of 2.2.2 (Table 6) and for the filter-style analysis (S20).

Why this dataset. The same designs as 2.2.2, so that the discrimination (active against the rest of the plate) and the ranking (among the actives) are two readings of one campaign. The question a designer asks of a metric is whether it would have enriched the hits on a plate, and this is a deposited design campaign on which that can be asked. The plate analyses of this subsection are **descriptive** and were planned before these data were analysed; the AUROC, added later, is reported in 2.2.2.

Metrics scored. The 36 items of Table 9 are those of Table 6: the 29 structure-space main-set metrics, the two prediction-based main-set metrics that the plates define (ipTM and the per-chain pTM minimum) and five reference rows (tyrosine fraction, net charge, sequence length, pLDDT and pTM). Designs are scored as deposited, and their predictions use the design's transition-state-analogue ligand and zinc, with accuracy measured against the design's own model. AME and the ligand clearance are in the Supplement (S21) and not among the ranked items, as in 2.2.2.

Ranking criteria. The correlation is Eq. (12) between the native value of the item on the design and $\log_{10}$ kcat/KM of the $n = 16$ designs that have a reported value; no direction is fixed in advance, so items are ranked by $|\rho_m|$ and the sign is reported ("higher value means higher or lower activity"). The interval is a 90% percentile bootstrap that resamples designs (2,000 draws). Beside it, the correlation is also computed with the sequence length partialled out of both sides (S22, S21): with the ranks r_x , r_y and r_z of the item, of the activity and of the length, centred, the residual of each rank on r_z is taken and the Pearson correlation of the residuals is the partial correlation,

$$e_x = r_x - \frac{r_z^\top r_x}{r_z^\top r_z} r_z, \quad e_y = r_y - \frac{r_z^\top r_y}{r_z^\top r_z} r_z, \quad \rho_{xy \cdot z} = \text{corr}(e_x, e_y). \quad (15)$$

A ρ is in bold in Table 9 when its 90% interval excludes 0, and about 10% of unrelated items are expected to show this by chance ($0.1 \times 36 = 3.6$ of the 36). The filter-style analysis fills a 96-well plate only from the quarter of the 192 designs that the item ranks highest, or lowest: with $n_{\text{keep}} = 48$ designs kept, of which k are active, the expected hits per plate are

$$\text{hits per plate} = 96 \frac{k}{n_{\text{keep}}} = 2k, \quad (16)$$

against $96 \times 16/192 = 8.0$ for an unfiltered plate. The band of a random filter is the distribution of k when 48 designs are chosen at random, hypergeometric with 16 active designs among 192,

$$\Pr(k \geq h) = \sum_{j \geq h} \frac{\binom{16}{j} \binom{176}{48-j}}{\binom{192}{48}}, \quad (17)$$

so that 95% of random choices hold at most 14.0 hits per plate ($h = 7$ actives), a random choice holds more with probability 0.022 and either end of a random item does with probability 0.044 (S20). The correlation of every structure-space metric and comparator, with and without the sequence length partialled out, is in Table S22, and the enrichment at keep fractions of 5, 10, 25 and 50% in both directions is in Table S23 (structure-space metrics) and Table S24 (prediction-based metrics).

2.5 Detection ability test: do the metrics respond more to a catalytic lesion than to a matched control?

Experiment. We damage the catalytic residues of natural enzymes, in the structure for the structure-space metrics and in the sequence of a re-predicted complex for the prediction-based metrics, and ask whether each metric moves more than it does when the same damage is made at a matched non-catalytic site of the same protein. We call each such deliberate change a *lesion*. In a serine protease, for example, the catalytic triad is Asp, His and Ser. The *Ala step* builds a structure in which those three residues are replaced by Ala. The *control* builds a structure in which three buried, non-catalytic residues of the same original types are replaced by Ala, with the same repacking. A metric is *specific* if it changes clearly more in the first structure than in the second. Two kinds of lesion are applied, each with its own datasets and table: the catalytic lesion (the chemistry ladder, 2.5.1; results in 3.4.1, Table 10) and the second-shell lesion (2.5.2; results in 3.4.2, Table 11).

Change. Every change is made in side-chain rotamer space on the fixed backbone, followed by an 8 Å fixed-backbone repack with the changed residues held and constrained minimisation (PyRosetta [21]); protonation is fixed. The **chemistry ladder** replaces each catalytic residue in one of four steps of increasing change in side-chain volume: **isosteric** (for example Ser → Cys, Asp → Asn, Thr → Val; median change 15.3 Å³), **non-isosteric** (36.4 Å³), **to Ala** (53.2 Å³) and **to Gly** (80.5 Å³). **Second-shell removal** mutates 1, 2 or 4 residues in the shell around the catalytic residues. The **detection floor** replaces all catalytic residues by Gly at once, which is the same lesion as the Gly step, and asks whether the metric notices. Table S25 gives the result of the detection floor and of the other sanity-floor conditions by class of metric, and checks that the detection floor and the Gly step are identical. For the prediction-based metrics the same four steps of the chemistry ladder are applied in the **sequence**: one catalytic residue is substituted per prediction and the complex is re-predicted with Chai-1 [13] (one seed, 101).

Comparator. Two comparisons are made, and they answer different questions.

- *Against the structure's own reference.* Every perturbed structure and its reference go through the identical treatment, because 26 of 103 metrics move under the treatment alone; each arm is compared with its own treated reference.
- *Against the matched control.* The control makes the **identical substitution** (same original and same new residue type) at buried, non-catalytic positions of the same protein, matched on burial (within 0.10 relative solvent-accessible area) and, in the primary analysis, on local packing, secondary structure and distance to the ligand where available. When no exact match exists, the matcher relaxes its criteria in a fixed order and records the loosest tier used (the loosest tier needed for each lesion test is in Table S26). Only comparisons where both arms change the same number of residues are counted (the balance of the numbers of residues changed in the two arms is in Table S27). For the re-predicted ladder each substitution is paired with the identical substitution at a buried non-catalytic position (same original and same new residue, matched on burial), and the change is measured against the unperturbed prediction of the same system at the same seed.

Response. For an item m , an enzyme i and an arm $a \in \{\text{cat}, \text{ctl}\}$ (the catalytic arm and the control arm), the response at step s is the change of the item from its own treated reference,

$$\Delta_i^a(m, s) = m(\text{changed}_{i,s}^a) - m(\text{reference}_i^a), \quad (18)$$

summarised as the median over enzymes. For the structure-space metrics the site-scoped items are computed over the changed residues in each arm (the catalytic residues in the catalytic arm, the control residues in the control arm); for second-shell removal both arms are scored over the catalytic residues. For the prediction-based metrics, both pair members are scored over the catalytic set *minus* the substituted position, so the active-site accuracy reports only the displacement of the remaining catalytic residues. An item **responds** at a step when the 95% interval of its median response excludes zero.

2.5.1 Detecting catalytic lesion

Dataset. The datasets of this test are listed in Table 4:

Table 4. Datasets of the catalytic lesion test. The sets on which the catalytic lesion is scored in Results 3.4.1, the sets added to them and the sets that are excluded or used for sensitivity analyses.

Dataset	Cases	How it was built, and what it is used for
M-CSA source [32]	434 curated entries (release 2026-07-31), 2,563 catalytic residues	the curated mechanisms and catalytic residues of the Mechanism and Catalytic Site Atlas; structures come from the Protein Data Bank [28, 29]
143-enzyme set (primary)	143 enzymes , 77 enzyme sub-subclasses, median 6 catalytic residues, 2 with a bound ligand	at most 25 per top-level EC class (25, 25, 25, 25, 25 and 18 enzymes), at least 2 resolved catalytic residues, ranked within a class by quality, catalytic-residue count and resolution; packing-matched, count-gated controls
Pilot set	53 enzymes (35 peptidases, 34 with a ligand), 52 added to the 143	active sites from UniProt [30] and M-CSA; one enzyme shares PDB entry 1LJL with a main enzyme and is counted once; controls matched on residue type and burial only
Structure-space evaluation set	195 enzymes (143 + 52)	the 29 structure-space main-set metrics; the main set alone is kept beside the pooled score (Table S28)
Prediction-based evaluation set	59 natural enzymes, 286 catalytic/control pairs	the natural arm of the re-predicted ladder (the 59 natural enzymes of 2.3, 28 of which are in the 143); the two reference rows pLDDT and pTM are scored on the same pairs
Excluded from pooling	30 de novo designs; 15 designed systems of the ladder (74 systems, 312 pairs with the 59 natural ones)	the control of a design is constructed, so designs are not pooled (Tables S29 and S30)
Shortened-motif set	the 143 enzymes with at most 3 catalytic residues kept	a sensitivity analysis of the specificity ratio (S31)

Why this dataset. A lesion at a catalytic residue is meaningful only for enzymes whose catalytic residues are curated, and M-CSA is the curated source. The 143-enzyme set was expanded from the 53-enzyme pilot for EC balance and not for N: the pilot was 66% peptidase, and with a flat result more systems buy precision on zero while broader coverage buys generality. The chemistry ladder isolates chemical identity at near-constant geometry, and its isosteric step is the hardest synthetic negative; the matched control, which is mandatory at every level, tests whether a metric knows that the lesion is at the catalytic site and not merely that a buried residue was changed. The pilot enzymes are pooled because the evaluation of a ranking needs as many enzymes as possible in each test; the effect of pooling is reported (adding the 52 pilot enzymes changes the mean R of a metric by at most 0.06, median 0.02, Table S28). The prediction-based arm uses the natural enzymes of the substrate-swap set, which have a cognate ligand, because the interface metrics are undefined without one.

Metrics scored. The 29 structure-space metrics (Table 1) are computed on every structure. The four prediction-based main-set metrics (Table 1, defined in 2.3) are computed on each re-predicted complex. The seed noise of the ladder is estimated from a replicate with a second seed on 30 natural pairs (Table S32). The 35 ranked items of Table 10 are the 29 structure-space and 4 prediction-based main-set metrics and the 2 reference rows (pLDDT, pTM); the 8 supplementary comparators and the per-step flags are in Table S28.

Ranking criteria. The **rank statistic R** is the probability form of the comparison between the two arms. It is the quantity that a shortlist depends on, because pipelines rank candidates and do not use magnitudes; Results 3.4 uses it to rank the metrics for each lesion. For an item m , a step s and the n_s enzymes that carry both arms and in which at least one arm moves,

$$R_{m,s} = \frac{1}{n_s} \sum_{i=1}^{n_s} \left[\mathbf{1}(|\Delta_i^{\text{cat}}| > |\Delta_i^{\text{ctl}}|) + \frac{1}{2} \mathbf{1}(|\Delta_i^{\text{cat}}| = |\Delta_i^{\text{ctl}}|) \right] \quad (19)$$

(an enzyme in which both changes are exactly zero carries no ranking information and is dropped; a step needs at least five enzymes). $R = 0.5$ means no specificity, $R > 0.5$ that the item moves more at the catalytic lesion, $R < 0.5$ that it moves more at the control; unlike a direction-free AUROC, it has a floor of 0 and not of 0.5.

Which steps count. Where the control arm does not move beyond the item's own noise, R measures whether the item moves at all and not whether it is specific, and where the catalytic response is not distinguishable from zero there is nothing to rank. A structure-space step therefore enters the set S_m of counted steps only if the item responds at that step (the interval of its median change excludes zero) and its control response is above its noise, as established on the 143 main enzymes. The prediction-based cells carry no such flag, because the control response of the predictor is noise-limited and is read through the distance-matched subset instead; they all count. The score of an item is the mean over the counted steps,

$$\bar{R}_m = \frac{1}{|S_m|} \sum_{s \in S_m} R_{m,s}, \quad (20)$$

and an item with no counted step has no mean. The number of steps counted for each item, $|S_m|$ out of the steps with a value, is in Table S28.

Interval. A 95% percentile bootstrap interval of $\bar{R}_m$ (Eq. 3 with $\bar{R}_m$ in place of A^* ; 5,000 draws) that resamples enzyme sub-classes, the first three EC fields, a stated proxy for superfamily, and for the predictor the systems; the same drawn clusters serve every step of an item.

Chance reference. As in 2.2, the top of a ranking of M items looks good by chance. The labels catalytic and control are swapped at random within each enzyme ($s_i^{(r)} \sim \text{Bernoulli}(1/2)$, $N_{\text{perm}} = 5,000$ replicates, the same swaps for every item), $\bar{R}_m^{(r)}$ is recomputed for every ranked item, and $T^{(r)} = \max_m \bar{R}_m^{(r)}$ (Eq. 4) has the distribution of the best score of M unrelated items; its median and 95th percentile are the **best-of-33 threshold** ($M = 33$ items that have a counted step; median 0.57, 95th percentile 0.62), a reference level for reading the ranking [37]. A mean above it is in bold in Table 10.

Onsets and grouping. The metrics are also differentiated by the lesion they can detect. The steps are ordered isosteric < non-isosteric < Ala < Gly (15.3, 36.4, 53.2 and 80.5 Å³). The **onset of detection** is the smallest step at which the item responds (Eq. 18; for the structure-space metrics the response analysis of the 143 main enzymes, 2,000 draws, seed 20260807; for the prediction-based metrics the signed changes of the natural pairs, bootstrap over systems),

$$s_{\text{det}}(m) = \min\{s : m \text{ responds at } s\}, \quad (21)$$

with "Never" if there is none. Rows of Table 10 are grouped by s_{det} (the first column), ranked by $\bar{R}_m$ within each group, and a metric with no counted step closes its group. The **onset of specificity** is the smallest step at which the item is specific under the specificity-ratio rule (structure-space metrics only; reported in Table S28); the ratio and the rule are

$$\text{SR}_{m,s} = \frac{\text{median}_i |\Delta_i^{\text{cat}}|}{\text{median}_i |\Delta_i^{\text{ctl}}|}, \quad (22)$$

and an item is called **specific** only when all five conditions hold: (1) the catalytic response is significant; (2) the interval of SR excludes 1; (3) the control arm contributes at least half as many observations as the catalytic arm; (4) the control response exceeds 1% of the item's own native standard deviation, because a ratio whose denominator is below the item's noise is not specificity; (5) both arms changed equal numbers of residues (2,000 draws, seed 20260807; counts are given as distinct metrics, with literal duplicates collapsed). The specificity ratio, its interval and the verdict (specific, non-specific, blind or invariant) of every eligible metric at every step are in Table S33, and those of the whole-protein and sequence-only comparators in Table S34; the 97 cells that survive the controls are listed in Table S35; and the per-step form of R on the 143 main enzymes alone, with 90% intervals and the comparators, is in Table S36.

Distance-matched R for the predictor. Catalytic residues touch the ligand by definition, so an interface item would respond more to a catalytic substitution than to a distant control even if it knew nothing about catalysis. The following plan was fixed before any distance was computed. For every pair the minimum distance d_{lig} from the substituted side chain to the ligand is measured, and R of Eq. (19) is also computed on the **caliper-matched subset**: pairs with the same residue types, the control not adjacent to a catalytic residue, and

$$|d_{\text{lig}}^{\text{cat}} - d_{\text{lig}}^{\text{ctl}}| \leq 2 \text{ Å}, \quad |\text{relSASA}^{\text{cat}} - \text{relSASA}^{\text{ctl}}| \leq 0.10, \quad (23)$$

extended by 54 pairs given a new in-caliper control (53 new predictions) wherever one existed, pooled over the four steps (68 to 69 pairs in 21 to 22 systems in Table 10). For the original control the median distance to the ligand was 13.4 Å against 3.1 Å for the catalytic residue. The distance of the substituted and the control residue to the ligand for every pair is in Table S37, the

candidate controls for the pairs that had none in the caliper, and why one was or was not selected, in Table S38, and the ratio in thirds of the distance to the ligand in Table S39.

The specificity-ratio analysis of the predictor (Table S30). The earlier analysis reports the ratio of the mean change at the catalytic residue to the mean change at the control (a median-based form is also given), with a bootstrap over systems (5,000 draws, seed 20260806), under three successively stricter controls: (i) the original control, on the isosteric step only; (ii) the caliper-matched subset of Eq. (23), with all four steps pooled (Table S40); (iii) a regression adjustment on all pairs and steps (Table S41),

$$\text{SR} = \frac{|\overline{\Delta}|_{\text{cat}}}{|\overline{\Delta}|_{\text{ctl}}}, \quad \log(|\Delta| + \varepsilon) = \beta_0 + \beta_{\text{arm}} \mathbf{1}(\text{catalytic}) + \beta_d d_{\text{lig}} + \beta_s \text{relSASA} + \text{error}, \quad (24)$$

where ε is a small metric-specific constant that keeps zero changes finite and the adjusted ratio is $\exp(\beta_{\text{arm}})$, with a bootstrap over systems. The **decision rule for the predictor** was fixed before the analysis: an interface item counts as specific only if the lower 90% bound of its mean-based SR in the matched subset exceeds 1.2, with at least 30 pairs in at least 15 systems; the active-site accuracy is reported as knock-on displacement with no specificity verdict, because the substituted residue is excluded from its own comparison. The whole structure-space panel is summarised in the Supplement by the geometric-mean SR over the metrics with a defined ratio, with a nested bootstrap (3,000 draws) that recomputes every metric inside each draw, so that correlation between metrics is carried through; equivalence to 1 is judged against a $\pm 25\%$ band (S42). The re-predicted ladder at each step, for all metrics, natural and de novo, is in Table S43.

2.5.2 Detecting second shell lesion

Dataset. The 143 main enzymes of 2.5.1 pooled with the 52 pilot enzymes that are not in the main set, with a control arm for up to 193 of them; structure-space metrics are scored on this lesion. The number of enzymes in which an item moves at all, and so carries information, ranges from 0 to 182 per item and dose.

Why this dataset. The second-shell lesion asks whether the metrics see past the first shell: second-shell residues around the catalytic residues are removed in graded doses of 1, 2 and 4 residues, and the control removes the same number of residues around a matched non-catalytic site. The same enzymes as the catalytic lesion are used, so that the two lesions can be read on one panel and one population, and the same pooling applies.

Metrics scored. The 29 structure-space main-set metrics, scored over the catalytic residues in both arms (so that a site-scoped item never knows which residues are catalytic). The 8 supplementary comparators and the per-dose flags are in Table S44.

Ranking criteria. The doses are $d \in \{1, 2, 4\}$ residues removed in place of the steps of Eq. (19), and the rank statistic is

$$R_{m,d} = \frac{1}{n_d} \sum_{i=1}^{n_d} \left[\mathbf{1}(|\Delta_i^{\text{cat}}| > |\Delta_i^{\text{ctl}}|) + \frac{1}{2} \mathbf{1}(|\Delta_i^{\text{cat}}| = |\Delta_i^{\text{ctl}}|) \right], \quad \bar{R}_m = \frac{1}{3} \sum_d R_{m,d}. \quad (25)$$

with the same bootstrap over enzyme sub-subclasses, the same onset of detection (Eq. 21, over doses) and the same grouping as for the catalytic lesion. The doses with a response and with a control response above the item's noise are counted for each item, and a dose enters a counted set only where both hold, as in 2.5.1; here no cell qualifies for any item, so the ranking uses the raw $R_{m,d}$ of every dose with at least five informative enzymes, raw values are reported, and no best-of- M threshold is used. Table 11 ranks the items within each detection group by raw $\bar{R}_m$ and gives, for each item, the number of doses (of 3) at which it responds and at which the control response is above its noise.

Provenance and data availability. Every number in this paper is read from a table generated by script from committed result files, with seeds and input hashes recorded. Each experiment's analysis plan was committed and dated before the analysis it governs, and amendments are listed with the date they were made. The Supplement gives, for every metric and every experiment, whether the metric was applicable and why; the full per-metric tables behind Tables 5–11; and the experiments that could not be made. All tables are released as CSV files (main-text tables in main/, supplementary tables in supplementary/); the Supplementary PDF describes every table and gives the path of its file. Structures and predictor weights are third-party and are not redistributed; their hashes are. BglB variant-level data are not redistributed because their licence is unstated; only aggregates are reported.

3 Results

Each subsection follows one order: the experiment, the dataset, the criteria by which the metrics are ranked, and the result table. Counts of metrics are of distinct metrics in the main set of 33 (29 structure-space, 4 prediction-based; Table 1 of the Methods), and intervals are 95% bootstrap intervals that resample the unit named in each subsection, unless stated otherwise. Numbers in square brackets are references, listed in the References. Complete per-metric tables and further analyses are in the supplementary tables (S numbers), which are described in the Supplementary PDF and released as CSV files (main/ and supplementary/).

3.1 Activity discrimination test

Experiment. Metrics are scored on real proteins in which one state is catalytically dead and the other active, and each metric is ranked by how well its value separates the two states (Methods 2.2). Two datasets are used, each with its own table.

3.1.1 Discriminate natural active and dead enzymes

Dataset. 21 zymogen–mature pairs from 21 peptidases (9 serine, 8 cysteine, 4 aspartic), downloaded from the RCSB Protein Data Bank [28, 29]. The zymogen is the dead form and the mature enzyme the active form, and the two forms share a ligand. Catalytic residues come from M-CSA [32] and UniProt [30] annotation; prediction-based quantities are Chai-1 [13] predictions of both forms with the shared ligand. Every item is scored on all 21 pairs (Methods 2.2.1).

Ranking criteria. The 33 main-set metrics and 6 reference rows are ranked by the direction-free AUROC [33, 34] of dead against active (0.5 = no separation), with a 95% bootstrap interval that resamples pairs. "Higher in" is the form in which the metric takes larger values. An AUROC in bold exceeds the best-of-39 threshold [37], the 95th percentile of the best AUROC among 39 unrelated items under random label swaps within pairs (median 0.63, 95th percentile 0.68).

Rank	Item	Group	Mode	AUROC [95% interval]	Higher in
1	<i>pLDDT</i> [12]	Reference row	prediction	0.79 [0.68, 0.90]	active
2	<i>pTM</i> [12]	Reference row	prediction	0.75 [0.62, 0.87]	active
3	Intra-residue repulsion (Rosetta, site) [20]	Rosetta energy	structure	0.61 [0.52, 0.72]	dead
4	Repulsion (Rosetta, site) [20]	Rosetta energy	structure	0.60 [0.51, 0.72]	dead
5	<i>Crystallographic resolution</i>	Reference row	structure	0.58 [0.50, 0.72]	dead
6	Fraction buried	Geometry	structure	0.57 [0.50, 0.69]	dead
7	Attraction (Rosetta, site) [20]	Rosetta energy	structure	0.56 [0.51, 0.64]	active
8	lk-ball solvation (Rosetta, site) [20]	Rosetta energy	structure	0.56 [0.50, 0.66]	dead
9	Ramachandran (Rosetta, site) [20]	Rosetta energy	structure	0.56 [0.51, 0.63]	dead
10	<i>Sequence length</i>	Reference row	structure	0.56 [0.50, 0.65]	dead
11	Relative SASA, mean	Geometry	structure	0.55 [0.50, 0.66]	active
12	Total SASA	Geometry	structure	0.55 [0.50, 0.67]	active
13	Rotamer, Dunbrack (Rosetta, site) [20]	Rosetta energy	structure	0.55 [0.50, 0.66]	dead
14	Pairwise distance, max	Geometry	structure	0.54 [0.50, 0.64]	dead
15	ipTM (Chai-1) [13, 27]	Interface confidence	prediction	0.54 [0.50, 0.61]	active
16	pKa, range (PROPKA) [25, 26]	pKa	structure	0.54 [0.50, 0.69]	dead
17	Polar contacts	Geometry	structure	0.54 [0.50, 0.59]	dead
18	Ligand clearance [8]	Active-site accuracy	prediction	0.54 [0.50, 0.65]	dead
19	Net van der Waals (Rosetta, site) [20]	Rosetta energy	structure	0.54 [0.50, 0.61]	active
20	Relative SASA, max	Geometry	structure	0.53 [0.50, 0.65]	active
21	Radius of gyration	Geometry	structure	0.53 [0.50, 0.64]	dead
22	AME RMSD [8]	Active-site accuracy	prediction	0.53 [0.50, 0.66]	active
23	Pairwise distance, mean	Geometry	structure	0.53 [0.50, 0.64]	dead
24	Solvation (Rosetta, site) [20]	Rosetta energy	structure	0.53 [0.50, 0.60]	dead
25	Omega torsion (Rosetta, site) [20]	Rosetta energy	structure	0.53 [0.50, 0.63]	dead
26	pKa, min (PROPKA) [25, 26]	pKa	structure	0.52 [0.50, 0.65]	active

Rank	Item	Group	Mode	AUROC [95% interval]	Higher in
27	Amino-acid propensity (Rosetta, site) [20]	Rosetta energy	structure	0.52 [0.50, 0.59]	active
28	Electrostatics (Rosetta, site) [20]	Rosetta energy	structure	0.52 [0.50, 0.58]	active
29	pKa, max (PROPKA) [25, 26]	pKa	structure	0.52 [0.50, 0.67]	dead
30	H-bond, backbone to side chain (Rosetta, site) [20]	Rosetta energy	structure	0.52 [0.50, 0.58]	dead

Table 5. Separation of real dead from active enzymes on the 21-pair evaluation set (top 30 of 39 ranked items). The 39 items are the 33 main-set metrics and the 6 reference rows (italic), ranked by the direction-free AUROC of dead against active (floor 0.5) over 21 pairs from 21 proteins; every row has $n = 21$ pairs. Intervals are 95% bootstrap intervals that resample pairs. "Higher in" is the form in which the metric takes larger values. An AUROC in bold exceeds the best-of-39 threshold, the 95th percentile of the best AUROC among 39 unrelated metrics in label-swap simulations (median 0.63, 95th percentile 0.68; defined in the Methods; a reference level for reading the ranking). The nine lowest-ranked items (AUROC 0.50 to 0.52) and all 158 metrics scored on these pairs are in Table S11. Source: main/T5_dead_vs_active.csv.

More detail. Every metric scored on the 21 pairs, 158 in all, with the best-of-158 threshold: Table S11. The wider set of 49 pairs (136 metrics) with the AUROC against the same-state null: Table S7. The definitions of the six pair sets and where each is used: Table S9. The 28 pairs re-predicted with Chai-1, restated with the definedness gate: Table S8. The check of the substrate-trapping mutants against the literature (26 of 30 not confirmed dead), a tier that was retired: Table S10. The reference rows in full, with intervals: Table S12.

3.1.2 Discriminate de novo designed active and no active enzymes

Dataset. 192 designs from a deposited metallohydrolase design campaign [3]: 16 have a reported kcat/KM (*active*) and 176 are *no active*. The comparison is active against no active designs. The designs lie in 136 backbone clusters, the 16 active designs in 8 of them (10 of the 16 in two clusters). Structure-space metrics are computed on the designs as deposited and prediction-based quantities on Chai-1 [13] predictions with the design's transition-state-analogue ligand and zinc. Every item is scored on all 192 designs (Methods 2.2.2).

Ranking criteria. The 36 items are 29 structure-space and 2 prediction-based main-set metrics and 5 reference rows (tyrosine fraction, net charge, sequence length, pLDDT, pTM). They are ranked by the direction-free AUROC [33, 34] of active against no active designs (0.5 = no separation), with a 95% bootstrap interval that resamples the 136 backbone clusters [36]. "Higher in" is the group in which the metric takes larger values. An AUROC in bold exceeds the best-of-36 threshold [37], the 95th percentile of the best AUROC among 36 items when the activity labels are permuted among designs (median 0.66, 95th percentile 0.73); the permutation treats designs as exchangeable, which understates the chance range when the actives are clumped in few clusters. AME is measured on the plates against each design's own model (a different metric from AME against a deposited structure) and is given in Table S13.

Rank	Item	Group	Mode	AUROC [95% interval]	Higher in
1	Intra-residue repulsion (Rosetta, site) [20]	Rosetta energy	structure	0.80 [0.64, 0.89]	no active
2	Repulsion (Rosetta, site) [20]	Rosetta energy	structure	0.78 [0.57, 0.92]	no active
3	<i>Tyrosine fraction</i>	Reference row	structure	0.75 [0.55, 0.86]	active
4	<i>pLDDT</i> [12]	Reference row	prediction	0.73 [0.53, 0.86]	active
5	<i>pTM</i> [12]	Reference row	prediction	0.72 [0.54, 0.85]	active
6	Site total (Rosetta) [20]	Rosetta energy	structure	0.71 [0.52, 0.82]	no active
7	Rotamer, Dunbrack (Rosetta, site) [20]	Rosetta energy	structure	0.71 [0.58, 0.88]	no active
8	Per-chain pTM minimum (Chai-1) [12, 13]	Interface confidence	prediction	0.70 [0.54, 0.83]	active
9	ipTM (Chai-1) [13, 27]	Interface confidence	prediction	0.70 [0.52, 0.84]	active
10	<i>Sequence length</i>	Reference row	structure	0.68 [0.51, 0.86]	no active
11	Worst residue (Rosetta, site) [20]	Rosetta energy	structure	0.68 [0.52, 0.78]	no active
12	Attraction (Rosetta, site) [20]	Rosetta energy	structure	0.67 [0.57, 0.82]	active
13	lk-ball solvation (Rosetta, site) [20]	Rosetta energy	structure	0.66 [0.54, 0.82]	active
14	Total SASA	Geometry	structure	0.65 [0.54, 0.89]	active
15	Relative SASA, mean	Geometry	structure	0.65 [0.53, 0.89]	active

Rank	Item	Group	Mode	AUROC [95% interval]	Higher in
16	Electrostatics (Rosetta, site) [20]	Rosetta energy	structure	0.64 [0.51, 0.79]	no active
17	Pairwise distance, max	Geometry	structure	0.64 [0.51, 0.77]	active
18	Relative SASA, max	Geometry	structure	0.62 [0.52, 0.84]	active
19	Pairwise distance, mean	Geometry	structure	0.61 [0.51, 0.77]	active
20	Radius of gyration	Geometry	structure	0.61 [0.51, 0.77]	active
21	H-bond, side chain (Rosetta, site) [20]	Rosetta energy	structure	0.61 [0.51, 0.73]	no active
22	Solvation (Rosetta, site) [20]	Rosetta energy	structure	0.60 [0.51, 0.71]	no active
23	Amino-acid propensity (Rosetta, site) [20]	Rosetta energy	structure	0.60 [0.51, 0.73]	active
24	Polar contacts	Geometry	structure	0.59 [0.50, 0.74]	active
25	Omega torsion (Rosetta, site) [20]	Rosetta energy	structure	0.58 [0.50, 0.81]	active
26	<i>Net charge</i>	Reference row	structure	0.56 [0.51, 0.83]	active
27	pKa, max (PROPKA) [25, 26]	pKa	structure	0.56 [0.50, 0.78]	no active
28	pKa, range (PROPKA) [25, 26]	pKa	structure	0.55 [0.50, 0.77]	no active
29	Net van der Waals (Rosetta, site) [20]	Rosetta energy	structure	0.54 [0.50, 0.80]	active
30	Fraction buried	Geometry	structure	0.53 [0.50, 0.70]	no active

Table 6. Separation of active from no active designs on the 192-design evaluation set (top 30 of 36 ranked items). The 36 items are 29 structure-space and 2 prediction-based main-set metrics and 5 reference rows (italic), ranked by the direction-free AUROC (floor 0.5) of the 16 active designs against the 176 no active designs; every row has $n = 192$ designs (16 active). Intervals are 95% bootstrap intervals that resample the 136 backbone clusters. "Higher in" is the group in which the metric takes larger values: "active" (the 16 designs with a reported kcat/KM) or "no active" (the other 176 designs). An AUROC in bold exceeds the best-of-36 threshold, the 95th percentile of the best AUROC among 36 unrelated items when the activity labels are permuted among designs (median 0.66, 95th percentile 0.73; a reference level for reading the ranking). Items 31 to 36 and all 128 metrics scored on the designs are in Table S13. Source: main/T6_plated_designs_auroc.csv.

More detail. Items 31 to 36 and every other metric scored on the 192 designs (128 in all), with the best-of-128 threshold: Table S13. The reference rows in full: Table S12. The descriptive ranking of the metrics among the 16 active designs is in 3.3.2.

3.2 Substrate discrimination test

Experiment. Each enzyme is predicted with Chai-1 [13] in four conditions that differ only in the ligand supplied: its cognate ligand, a wrong ligand of the same enzyme class, a wrong ligand of a different class, and a decoy ligand taken from an unrelated structure. Each prediction-based quantity is ranked by how well it separates the cognate prediction from a wrong-ligand prediction of the same enzyme (Methods 2.3).

Dataset. 55 natural enzymes of M-CSA [32], of the 59 in the experiment: on these, every scored quantity has a defined value in all four conditions and every supplied ligand was present in the prediction (four enzymes are excluded because a heme-type ligand could not be built into a chemical structure and the predictor dropped it; the analysis on all 59 is in Table S14). One n , 55 enzymes, applies to every row (Methods 2.3).

Ranking criteria. Chai-1 returns five models per prediction, and each score is summarised over them as a mean, a maximum or a spread (standard deviation across models), and for the AME RMSD also as a minimum, so one metric appears in several rows. The 30 scored quantities are the four prediction-based main-set metrics (ipTM, per-chain pTM minimum, ligand clearance, AME RMSD), whole-prediction confidence scores (including the reference rows pLDDT and pTM), the AME pass bit and four bookkeeping counters. Each is ranked by the mean of three direction-free AUROCs [33, 34], one per kind of wrong ligand (same class, different class, decoy), each with a 95% interval that resamples enzymes. "Higher in" is the condition in which the quantity takes larger values. A mean in bold exceeds the best-of-30 threshold [37], the 95th percentile of the best AUROC among 30 quantities when the four ligand conditions are permuted within each enzyme (median 0.56, 95th percentile 0.61).

Rank	Quantity	Summary over the five models	Group	Same class	Different class	Decoy	Mean	Higher in
1	ipTM [13, 27]	mean	Interface confidence (main set)	0.69 [0.60, 0.78]	0.75 [0.66, 0.84]	0.65 [0.55, 0.76]	0.70	cognate

Rank	Quantity	Summary over the five models	Group	Same class	Different class	Decoy	Mean	Higher in
2	ipTM [13, 27]	maximum	Interface confidence (main set)	0.68 [0.59, 0.77]	0.74 [0.65, 0.83]	0.65 [0.54, 0.74]	0.69	cognate
3	ipTM [13, 27]	spread (SD)	Interface confidence (main set)	0.70 [0.60, 0.79]	0.69 [0.60, 0.77]	0.67 [0.59, 0.75]	0.69	wrong ligand
4	pTM [12]	spread (SD)	Whole-prediction confidence	0.62 [0.55, 0.70]	0.64 [0.58, 0.71]	0.64 [0.57, 0.71]	0.64	wrong ligand
5	Ligand clearance [8]	best model	Active-site accuracy (main set)	0.61 [0.52, 0.69]	0.65 [0.57, 0.74]	0.62 [0.53, 0.71]	0.63	wrong ligand
6	pTM [12]	mean	Reference row	0.60 [0.56, 0.66]	0.63 [0.58, 0.69]	0.64 [0.58, 0.70]	0.62	cognate
7	pTM [12]	maximum	Whole-prediction confidence	0.60 [0.56, 0.66]	0.63 [0.58, 0.69]	0.63 [0.58, 0.70]	0.62	cognate
8	Per-chain pTM minimum [12, 13]	spread (SD)	Interface confidence (main set)	0.58 [0.51, 0.68]	0.58 [0.51, 0.68]	0.69 [0.59, 0.79]	0.62	wrong ligand
9	AME RMSD [8]	spread (SD)	Active-site accuracy (main set)	0.55 [0.51, 0.60]	0.56 [0.51, 0.62]	0.57 [0.52, 0.63]	0.56	wrong ligand
10	Per-chain pTM minimum [12, 13]	mean	Interface confidence (main set)	0.58 [0.51, 0.67]	0.60 [0.51, 0.71]	0.50 [0.50, 0.62]	0.56	cognate
11	Per-chain pTM minimum [12, 13]	maximum	Interface confidence (main set)	0.58 [0.51, 0.67]	0.59 [0.51, 0.71]	0.50 [0.50, 0.62]	0.56	cognate
12	AME RMSD [8]	maximum	Active-site accuracy (main set)	0.53 [0.50, 0.57]	0.54 [0.51, 0.58]	0.54 [0.51, 0.59]	0.54	wrong ligand
13	AME RMSD [8]	minimum	Active-site accuracy (main set)	0.53 [0.50, 0.56]	0.54 [0.51, 0.57]	0.52 [0.50, 0.57]	0.53	wrong ligand
14	AME RMSD, no symmetry resolution [8]	minimum	Active-site accuracy (main set)	0.52 [0.50, 0.56]	0.54 [0.51, 0.57]	0.53 [0.50, 0.57]	0.53	wrong ligand
15	AME RMSD [8]	mean	Active-site accuracy (main set)	0.53 [0.50, 0.56]	0.53 [0.51, 0.57]	0.52 [0.50, 0.57]	0.53	wrong ligand
16	pLDDT [12]	spread (SD)	Whole-prediction confidence	0.51 [0.50, 0.55]	0.52 [0.50, 0.57]	0.54 [0.50, 0.60]	0.52	mixed
17	Per-chain pTM of the protein [12, 13]	mean	Whole-prediction confidence	0.52 [0.50, 0.54]	0.52 [0.50, 0.54]	0.53 [0.51, 0.56]	0.52	cognate
18	Per-chain pTM of the protein [12, 13]	maximum	Whole-prediction confidence	0.52 [0.50, 0.54]	0.52 [0.50, 0.55]	0.53 [0.51, 0.55]	0.52	cognate
19	pLDDT [12]	best model	Whole-prediction confidence	0.52 [0.50, 0.54]	0.53 [0.51, 0.55]	0.51 [0.50, 0.54]	0.52	cognate
20	pLDDT [12]	mean	Reference row	0.52 [0.50, 0.54]	0.53 [0.51, 0.55]	0.51 [0.50, 0.54]	0.52	cognate
21	Per-chain pTM of the protein [12, 13]	spread (SD)	Whole-prediction confidence	0.50 [0.50, 0.55]	0.52 [0.50, 0.56]	0.50 [0.50, 0.55]	0.51	wrong ligand
22	Symmetry swaps (count)	single value	Bookkeeping counter	0.51 [0.50, 0.54]	0.51 [0.50, 0.53]	0.50 [0.50, 0.52]	0.51	cognate
23	pLDDT [12]	lowest residue	Whole-prediction confidence	0.50 [0.50, 0.52]	0.50 [0.50, 0.53]	0.50 [0.50, 0.53]	0.50	cognate
24	AME pass bit [8]	single value	Active-site accuracy (pass bit)	0.50 [0.50, 0.50]	0.50 [0.50, 0.53]	0.50 [0.50, 0.53]	0.50	mixed
25	Backbone atoms aligned (count)	single value	Bookkeeping counter	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50	mixed
26	Catalytic atoms (count)	single value	Bookkeeping counter	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50	mixed
27	Models predicted (count)	single value	Bookkeeping counter	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50	mixed
28	Inter-chain clash flag [13]	maximum	Whole-prediction confidence	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50	mixed
29	Inter-chain clash flag [13]	mean	Whole-prediction confidence	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50	mixed
30	Inter-chain clash flag [13]	spread (SD)	Whole-prediction confidence	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50	mixed

Table 7. The substrate swap on the 55-enzyme evaluation set, 30 scored prediction-based quantities ranked by AUROC. Direction-free AUROC (floor 0.5) of the cognate prediction against a wrong-ligand prediction of the same enzyme, in three columns by kind of wrong ligand, with 95% intervals that resample enzymes; the value of an enzyme in a condition is the mean over seeds 101, 202 and 303. Rows are ranked by the mean of the three AUROCs; a mean in bold exceeds the best-of-30 threshold (median 0.56, 95th percentile 0.61; permutation of the four ligand conditions within each enzyme, defined in the Methods; a reference level for reading the ranking). Chai-1 predicts five models for every prediction, and each score is summarised over them (mean, maximum, spread as the standard deviation across models, and for the AME RMSD also the minimum), so one metric appears in several rows, one per summary. Reference rows are in italic. Source: main/T7_substrate_swap.csv.

More detail. The analysis on all 59 natural enzymes (five-seed cognate mean) and on the 30 de novo designs, for all 33 scored prediction-based quantities including the Chai-1 combined score, with the paired change in units of seed noise: Table S14. The same comparison stratified by the difference in ligand size and charge, with paired win fractions: Table S15. The drift of the current Chai-1 engine against the earlier predictions on a re-run shard: Table S45.

3.3 Activity ranking test

Experiment. Metrics are scored on real proteins whose activity was measured, and each metric is ranked by the rank correlation between its value and the measured activity (Methods 2.4). Two datasets are used, each with its own table.

3.3.1 Ranking natural enzyme activity

Dataset. BglB, a β -glucosidase from *Paenibacillus polymyxa*, with single-point variants characterised by the D2DCure consortium [38] (data retrieved from d2dcure.com on 2026-08-06): 655 variants at 207 positions, of which the **432 variants at 175 positions that are folded (expressed and not destabilised) and have kinetics** are the evaluation set. The measured value is the *impairment*, the log10 of the loss of kcat/KM relative to the wild type (larger = more impaired). Each variant is built in the computer on the crystal structure of the wild type (RCSB PDB entry 2JIE [28, 29], which carries a covalent glucosyl intermediate) and structure-space metrics are computed over the two catalytic glutamates (Glu167 and Glu356, from the UniProt [30] active-site features) before and after the substitution; prediction-based metrics are computed on Chai-1 [13] predictions of the variant with the assay substrate (pNPG) (Methods 2.4.1).

Ranking criteria. The 41 items are the 28 structure-space main-set metrics that BglB defines (the omega term does not change under a side-chain substitution on a fixed backbone), the 4 prediction-based main-set metrics, 4 reference rows (tyrosine fraction, net charge, pLDDT, pTM) and 5 baselines declared before any metric was joined to a label: the distance from the mutated residue to the nearest catalytic residue, the burial of the mutated residue, the BLOSUM62 score of the substitution, the change in side-chain volume and the Rosetta score that comes with the BglB data (248 of the 432 variants). Each is ranked by the Spearman rank correlation ρ [41] with the measured impairment, with a 95% bootstrap interval that resamples the 175 positions [36]. For a metric the score is the size of its change (the absolute difference between the variant and the unsubstituted protein), with the direction fixed in advance (larger in more impaired variants); for a baseline it is the baseline itself, pointed so that a larger value means more impaired. A ρ in bold exceeds the best-of-41 threshold [37], the 95th percentile of the best ρ among 41 items when the measured impairments are permuted among whole positions (median 0.24, 95th percentile 0.32). The last column says whether the item beats the distance baseline: the interval of the difference between its ρ and the baseline's, on the same bootstrap draws, lies above 0.

Rank	Item	Group	Mode	Spearman ρ [95% interval]	Beats the distance baseline
1	pKa, range (PROPKA) [25, 26]	pKa	structure	0.48 [0.35, 0.59]	no
2	pKa, min (PROPKA) [25, 26]	pKa	structure	0.46 [0.32, 0.57]	no
3	pKa shift, max absolute (PROPKA) [25, 26]	pKa	structure	0.46 [0.32, 0.58]	no
4	pKa, max (PROPKA) [25, 26]	pKa	structure	0.46 [0.32, 0.58]	no
5	<i>Distance to the catalytic residues</i>	Reference row	—	0.46 [0.32, 0.57]	—
6	Intra-residue repulsion (Rosetta, site) [20]	Rosetta energy	structure	0.45 [0.30, 0.57]	no
7	Relative SASA, mean	Geometry	structure	0.45 [0.32, 0.57]	no
8	pKa shift, mean (PROPKA) [25, 26]	pKa	structure	0.45 [0.31, 0.57]	no
9	pKa, mean (PROPKA) [25, 26]	pKa	structure	0.45 [0.31, 0.57]	no
10	Total SASA	Geometry	structure	0.45 [0.31, 0.57]	no
11	Repulsion (Rosetta, site) [20]	Rosetta energy	structure	0.45 [0.31, 0.57]	no

Rank	Item	Group	Mode	Spearman ρ [95% interval]	Beats the distance baseline
12	Pairwise distance, max	Geometry	structure	0.45 [0.30, 0.56]	no
13	Attraction (Rosetta, site) [20]	Rosetta energy	structure	0.44 [0.30, 0.56]	no
14	Rotamer, Dunbrack (Rosetta, site) [20]	Rosetta energy	structure	0.44 [0.29, 0.56]	no
15	lk-ball solvation (Rosetta, site) [20]	Rosetta energy	structure	0.44 [0.31, 0.54]	no
16	Radius of gyration	Geometry	structure	0.43 [0.28, 0.56]	no
17	Net van der Waals (Rosetta, site) [20]	Rosetta energy	structure	0.43 [0.28, 0.55]	no
18	Relative SASA, max	Geometry	structure	0.43 [0.29, 0.54]	no
19	Pairwise distance, mean	Geometry	structure	0.43 [0.28, 0.55]	no
20	Electrostatics (Rosetta, site) [20]	Rosetta energy	structure	0.43 [0.28, 0.55]	no
21	Site total (Rosetta) [20]	Rosetta energy	structure	0.42 [0.27, 0.55]	no
22	Worst residue (Rosetta, site) [20]	Rosetta energy	structure	0.42 [0.26, 0.55]	no
23	Solvation (Rosetta, site) [20]	Rosetta energy	structure	0.41 [0.27, 0.54]	no
24	H-bond, side chain (Rosetta, site) [20]	Rosetta energy	structure	0.35 [0.19, 0.49]	no
25	<i>Burial of the mutated residue</i>	Reference row	–	0.33 [0.21, 0.43]	–
26	ipTM (Chai-1) [13, 27]	Interface confidence	prediction	0.33 [0.19, 0.45]	no
27	Per-chain pTM minimum (Chai-1) [12, 13]	Interface confidence	prediction	0.28 [0.15, 0.41]	no
28	<i>pTM</i> [12]	Reference row	prediction	0.26 [0.12, 0.38]	no
29	<i>Rosetta score in the BglB data set</i>	Reference row	–	0.21 [0.06, 0.35]	–
30	Fraction buried	Geometry	structure	0.19 [0.06, 0.30]	no

Table 8. BglB variants: top 30 of 41 ranked items on the 432-variant evaluation set. Spearman rank correlation ρ between each item and the measured impairment of the 432 variants, with a 95% interval that resamples the 175 positions. For a metric the score is the size of its change, for a baseline its own score, with the direction fixed in advance (a larger change in more impaired variants); a negative ρ is reported as it is. The 41 items are the 32 main-set metrics that BglB defines, four reference rows and five declared baselines; reference rows and baselines are in italic. A value in bold exceeds the best-of-41 threshold (median 0.24, 95th percentile 0.32; permutation of the measured impairments among whole positions; a reference level for reading the ranking). Structure-space rows have 431 variants with a value, prediction-based rows 432 and the Rosetta score of the BglB data 248. Items 31 to 41, and all 114 quantities analysed, are in Table S16. Source: main/T8_bglb_variants.csv.

More detail. All 114 quantities analysed, ranked, with the AUROC of the binary contrast (impaired against wild-type-like) beside each rank correlation: Table S16. The 84 structure-space and comparator metrics with the signed, direction-free and magnitude AUROC and the comparison with every baseline: Table S17. The 28 prediction-based metrics: Table S19. The main analysis and its ten sensitivity analyses: Table S18. The five declared baselines and the reference rows in full: Table S12.

3.3.2 Ranking de novo designed enzyme activity

Dataset. The 192-design campaign of 3.1.2 [3]. The 16 designs with a reported kcat/KM have a measured activity and form the evaluation set, so every row has $n = 16$; they lie in 8 of the 136 backbone clusters (Methods 2.4.2).

Ranking criteria. The 36 items are those of Table 6. Each is ranked by the size of the Spearman correlation ρ [41] between its value on the design as deposited and $\log_{10}$ kcat/KM, with a 90% bootstrap interval over designs. The sign of ρ is reported ("higher value means higher or lower activity"), and the last column gives ρ with the sequence length partialled out of both sides. A ρ in bold has an interval that excludes 0, which about 3.6 of the 36 items would show for unrelated metrics. The analysis is descriptive (Methods 2.4.2).

Rank	Item	Group	Mode	Spearman ρ [90% interval]	Higher value means	ρ , length partialled out
1	Polar contacts	Geometry	structure	0.67 [0.41, 0.85]	higher activity	0.52
2	Net van der Waals (Rosetta, site) [20]	Rosetta energy	structure	-0.63 [-0.88, -0.24]	lower activity	-0.47
3	<i>pLDDT</i> [12]	Reference row	prediction	0.60 [0.24, 0.83]	higher activity	0.36
4	<i>Sequence length</i>	Reference row	structure	-0.58 [-0.82, -0.23]	lower activity	–
5	Solvation (Rosetta, site) [20]	Rosetta energy	structure	0.57 [0.15, 0.85]	higher activity	0.54

Rank	Item	Group	Mode	Spearman ρ [90% interval]	Higher value means	ρ , length partialled out
6	Amino-acid propensity (Rosetta, site) [20]	Rosetta energy	structure	0.54 [0.08, 0.81]	higher activity	0.64
7	<i>Net charge</i>	Reference row	structure	0.46 [−0.02, 0.80]	higher activity	0.22
8	Pairwise distance, mean	Geometry	structure	−0.43 [−0.71, 0.02]	lower activity	−0.29
9	Site total (Rosetta) [20]	Rosetta energy	structure	−0.42 [−0.69, 0.01]	lower activity	−0.37
10	Radius of gyration	Geometry	structure	−0.41 [−0.69, 0.03]	lower activity	−0.29
11	H-bond, side chain (Rosetta, site) [20]	Rosetta energy	structure	−0.40 [−0.74, 0.05]	lower activity	−0.14
12	Ramachandran (Rosetta, site) [20]	Rosetta energy	structure	−0.38 [−0.82, 0.15]	lower activity	−0.09
13	Repulsion (Rosetta, site) [20]	Rosetta energy	structure	−0.33 [−0.73, 0.17]	lower activity	0.21
14	pKa, mean (PROPKA) [25, 26]	pKa	structure	−0.30 [−0.57, 0.09]	lower activity	0.09
15	pKa shift, mean (PROPKA) [25, 26]	pKa	structure	−0.30 [−0.56, 0.09]	lower activity	0.09
16	Relative SASA, mean	Geometry	structure	−0.25 [−0.76, 0.31]	lower activity	−0.35
17	Total SASA	Geometry	structure	−0.25 [−0.76, 0.30]	lower activity	−0.35
18	Relative SASA, max	Geometry	structure	−0.25 [−0.76, 0.28]	lower activity	−0.34
19	Attraction (Rosetta, site) [20]	Rosetta energy	structure	−0.25 [−0.66, 0.25]	lower activity	−0.20
20	Pairwise distance, max	Geometry	structure	0.23 [−0.26, 0.64]	higher activity	0.26
21	Intra-residue repulsion (Rosetta, site) [20]	Rosetta energy	structure	−0.23 [−0.65, 0.30]	lower activity	−0.46
22	Ik-ball solvation (Rosetta, site) [20]	Rosetta energy	structure	−0.21 [−0.53, 0.19]	lower activity	0.08
23	Electrostatics (Rosetta, site) [20]	Rosetta energy	structure	−0.19 [−0.58, 0.25]	lower activity	−0.61
24	Per-chain pTM minimum (Chai-1) [12, 13]	Interface confidence	prediction	0.18 [−0.35, 0.66]	higher activity	−0.00
25	Rotamer, Dunbrack (Rosetta, site) [20]	Rosetta energy	structure	0.18 [−0.31, 0.57]	higher activity	0.11
26	H-bond, backbone to side chain (Rosetta, site) [20]	Rosetta energy	structure	0.17 [−0.21, 0.52]	higher activity	0.32
27	pKa, max (PROPKA) [25, 26]	pKa	structure	−0.16 [−0.49, 0.26]	lower activity	0.16
28	<i>pTM</i> [12]	Reference row	prediction	−0.15 [−0.64, 0.41]	lower activity	−0.26
29	<i>Tyrosine fraction</i>	Reference row	structure	−0.14 [−0.68, 0.40]	lower activity	0.00
30	pKa, range (PROPKA) [25, 26]	pKa	structure	−0.13 [−0.53, 0.31]	lower activity	−0.07

Table 9. Plated designs: top 30 of 36 ranked items on the 16 designs that have a reported kcat/KM. Spearman rank correlation ρ between each metric (native value on the design) and $\log_{10}$ kcat/KM, with a 90% interval that resamples designs; items are ranked by the size of ρ . A value in bold has an interval that excludes 0 (about 3.6 of the 36 items would show this for unrelated metrics). The last column gives ρ with the sequence length partialled out of both sides. The 36 items are 29 structure-space and 2 prediction-based main-set metrics and 5 reference rows (italic). Descriptive analysis. Items 31 to 36, and all 106 metrics with a correlation, are in Table S21; the hits-per-plate analysis of the 192 designs is in Table S20. Source: main/T9_plated_designs.csv.

More detail. The correlation with kcat/KM of all 106 metrics that have one, with the length control: Table S21. The correlations of the structure-space metrics and comparators, with and without the length control: Table S22. The hits per plate of all 128 metrics when each fills a 96-well plate from the quarter of the 192 designs that it ranks highest or lowest: Table S20. The enrichment at keep fractions of 5, 10, 25 and 50% in both directions: Table S23 (102 structure-space metrics) and Table S24 (29 prediction-based metrics).

3.4 Detection ability test

Experiment. Each metric is ranked by whether it changes more under a lesion at a catalytic residue than under the identical lesion at a matched non-catalytic control position of the same protein (Methods 2.5). Two kinds of lesion are tested, each with its own datasets and table: the catalytic lesion (3.4.1) and the second-shell lesion (3.4.2).

3.4.1 Detecting catalytic lesion

Experiment. One catalytic residue is replaced in four steps of increasing change in side-chain volume (isosteric 15 Å³, non-isosteric 36 Å³, to Ala 53 Å³, to Gly 81 Å³), in the crystal structure by a rotamer-space edit and repack, or in the sequence of the re-predicted complex with Chai-1 [13]; the control makes the identical substitution at a matched buried non-catalytic position of the same protein.

Dataset. For the 29 structure-space main-set metrics: the 143 M-CSA enzymes [32] (77 enzyme sub-subclasses; packing-matched, count-gated controls) pooled with the 52 enzymes of the 53-enzyme pilot set that are not in it (active sites from UniProt [30] and M-CSA; controls matched on residue type and burial only; the one pilot enzyme that shares a PDB entry with the main set is counted once), 195 enzymes in all. For the four prediction-based main-set metrics and the two reference rows (pLDDT and pTM): the natural arm of the re-predicted ladder, the 59 natural enzymes of the substrate-swap set with 286 catalytic/control pairs. The 30 de novo designs and the 15 designed systems of the ladder, whose control is constructed, are analysed separately (Tables S29 and S30) (Methods 2.5.1).

Ranking criteria. The 35 items are the 29 structure-space and 4 prediction-based main-set metrics and the 2 reference rows (italic). The rank statistic R is the probability that a metric changes more at the catalytic lesion than at the matched control of the same protein (0.5 = no specificity; above 0.5, more change at the catalytic lesion). The change of a metric is the absolute difference between the perturbed and the unperturbed structure (or prediction), a tie counts one half, and proteins in which neither arm moves are dropped. A step counts toward a structure-space metric only if the metric responds to the lesion (the interval of its median change excludes zero) and the control response is above the metric's own noise; a step that is not counted is in parentheses, and "-" is a step with fewer than five informative proteins. The mean R over the counted steps has a 95% bootstrap interval that resamples enzyme sub-subclasses (systems for the predictor). Rows are grouped by the lesion at which the metric is first detected (the smallest step at which it responds; "Never" if at none) and ranked by mean R within each group; a metric with no counted step has no mean and closes its group. A mean in bold exceeds the best-of-33 threshold [37], the 95th percentile of the best R among 33 items when the catalytic and control labels are swapped within each protein (median 0.57, 95th percentile 0.62). The distance-matched R (prediction-based metrics only) is computed on the pairs whose control lies at the same distance from the ligand as the catalytic residue (within 2 Å and within 0.10 in relative solvent-accessible area), pooled over the four steps (68 to 69 pairs in 21 to 22 systems); the original control of the predictor sat a median 13.4 Å from the ligand, the catalytic residue 3.1 Å (the distances for every pair are in Table S37 and the caliper-matched subsets in Table S40). The number of steps counted for each metric and the smallest step at which a structure-space metric is specific (it responds, the specificity ratio is above 1, the control response is above the noise and both arms changed equal numbers of residues) are in Table S28.

Detected from	Rank in group	Item	R at isosteric / non-isosteric / Ala / Gly	Mean R [95% interval]	R, distance-matched [95% interval]
Isosteric	1	Per-chain pTM minimum (Chai-1) [12, 13]	0.81 / 0.73 / 0.75 / 0.72	0.75 [0.68, 0.82]	0.59 [0.42, 0.76]
Isosteric	2	pKa, range (PROPKA) [25, 26]	0.70 / 0.67 / - / -	0.68 [0.58, 0.78]	-
Isosteric	3	ipTM (Chai-1) [13, 27]	0.71 / 0.71 / 0.69 / 0.58	0.67 [0.60, 0.75]	0.35 [0.24, 0.50]
Isosteric	4	Solvation (Rosetta, site) [20]	0.59 / 0.73 / 0.60 / 0.62	0.63 [0.57, 0.70]	-
Isosteric	5	Relative SASA, mean	0.48 / 0.55 / 0.73 / 0.69	0.61 [0.56, 0.67]	-
Isosteric	6	pKa, min (PROPKA) [25, 26]	0.54 / 0.67 / - / -	0.61 [0.53, 0.67]	-
Isosteric	7	Repulsion (Rosetta, site) [20]	0.48 / 0.55 / 0.64 / 0.62	0.57 [0.52, 0.62]	-
Isosteric	8	pKa shift, mean (PROPKA) [25, 26]	0.46 / 0.67 / - / -	0.56 [0.47, 0.66]	-
Isosteric	9	pTM [12]	0.60 / 0.48 / 0.60 / 0.58	0.56 [0.49, 0.64]	0.45 [0.30, 0.60]
Isosteric	10	Relative SASA, max	0.49 / (0.50) / 0.56 / 0.55	0.53 [0.48, 0.59]	-
Isosteric	11	Attraction (Rosetta, site) [20]	0.53 / 0.58 / 0.55 / 0.46	0.53 [0.46, 0.60]	-
Isosteric	12	pLDDT [12]	0.62 / 0.37 / 0.55 / 0.58	0.53 [0.46, 0.60]	0.49 [0.34, 0.63]
Isosteric	13	Ramachandran (Rosetta, site) [20]	0.43 / 0.55 / 0.55 / 0.56	0.52 [0.47, 0.59]	-
Isosteric	14	Intra-residue repulsion (Rosetta, site) [20]	0.48 / 0.59 / 0.48 / 0.52	0.52 [0.47, 0.57]	-
Isosteric	15	Rotamer, Dunbrack (Rosetta, site) [20]	0.45 / 0.47 / 0.58 / 0.58	0.52 [0.46, 0.57]	-
Isosteric	16	Radius of gyration	0.42 / 0.48 / 0.62 / -	0.51 [0.45, 0.57]	-

Detected from	Rank in group	Item	R at isosteric / non-isosteric / Ala / Gly	Mean R [95% interval]	R, distance-matched [95% interval]
Isosteric	17	Amino-acid propensity (Rosetta, site) [20]	0.45 / (0.57) / 0.52 / 0.53	0.50 [0.45, 0.55]	–
Isosteric	18	pKa, max (PROPKA) [25, 26]	0.43 / 0.56 / – / –	0.49 [0.39, 0.61]	–
Isosteric	19	Worst residue (Rosetta, site) [20]	0.45 / 0.46 / 0.54 / (0.43)	0.48 [0.44, 0.53]	–
Isosteric	20	Net van der Waals (Rosetta, site) [20]	0.45 / 0.55 / 0.47 / 0.39	0.46 [0.39, 0.52]	–
Isosteric	21	H-bond, side chain (Rosetta, site) [20]	0.47 / 0.42 / 0.43 / 0.42	0.43 [0.35, 0.52]	–
Isosteric	22	Site total (Rosetta) [20]	0.41 / 0.58 / 0.41 / 0.29	0.42 [0.37, 0.47]	–
Isosteric	23	Pairwise distance, mean	0.43 / 0.49 / 0.32 / –	0.41 [0.34, 0.50]	–
Isosteric	24	pKa, mean (PROPKA) [25, 26]	0.40 / (0.47) / – / –	0.40 [0.28, 0.54]	–
Isosteric	25	H-bond, backbone to side chain (Rosetta, site) [20]	0.26 / 0.29 / 0.31 / 0.29	0.29 [0.19, 0.39]	–
Isosteric	26	Electrostatics (Rosetta, site) [20]	0.26 / 0.20 / 0.24 / 0.26	0.24 [0.18, 0.30]	–
Isosteric	–	Polar contacts	(0.93) / (0.97) / (0.96) / (0.96)	–	–
Non-isosteric	1	AME RMSD [8]	0.61 / 0.73 / 0.68 / 0.72	0.68 [0.62, 0.75]	0.45 [0.33, 0.57]
Non-isosteric	2	lk-ball solvation (Rosetta, site) [20]	(0.56) / 0.53 / 0.59 / 0.65	0.59 [0.51, 0.67]	–
Non-isosteric	3	Total SASA	(0.46) / 0.51 / (0.49) / 0.58	0.55 [0.48, 0.60]	–
Non-isosteric	4	pKa shift, max absolute (PROPKA) [25, 26]	(0.53) / 0.53 / – / –	0.53 [0.42, 0.65]	–
Ala	1	Fraction buried	(0.54) / (0.56) / 0.70 / 0.63	0.66 [0.59, 0.74]	–
Gly	1	Omega torsion (Rosetta, site) [20]	(0.46) / – / (0.58) / 0.52	0.52 [0.45, 0.60]	–
Never	1	Ligand clearance [8]	0.57 / 0.63 / 0.57 / 0.63	0.60 [0.54, 0.66]	0.57 [0.44, 0.70]
Never	–	Pairwise distance, max	(0.30) / (0.36) / (0.19) / –	–	–

Table 10. Catalytic lesion (chemistry ladder): the 35 ranked items grouped by the lesion at which they are first detected and ranked by R within the group. Rank statistic R, the probability that a metric changes more at the catalytic lesion than at the matched control (0.5 = no specificity), at the four steps and as the mean over the steps counted, with a 95% interval that resamples enzyme sub-subclasses (systems for the predictor). "Detected from" is the smallest lesion step (isosteric 15 Å³, non-isosteric 36 Å³, Ala 53 Å³, Gly 81 Å³) at which the metric responds. Structure-space metrics: up to 195 natural enzymes (143 main + 52 pilot; the number of enzymes in which a metric moves at all ranges from 1 to 150 per metric and step); prediction-based metrics (ipTM, per-chain pTM minimum, ligand clearance, AME RMSD) and the reference rows pLDDT and pTM (italic): 59 natural enzymes, 286 pairs. A step in parentheses is not counted. A mean in bold exceeds the best-of-33 threshold (median 0.57, 95th percentile 0.62; a reference level for reading the ranking). The distance-matched R is given for the prediction-based metrics. The 8 supplementary comparators, the number of steps counted, the onset of specificity and all per-step flags are in Table S28. Source: main/T10_specificity_catalytic_lesion.csv.

More detail.

- *Ranking.* All 43 metrics scored for the catalytic lesion, with the 8 supplementary comparators, R and the response, control-noise and specificity flags at each step, and the mean R of the 143 main enzymes alone: Table S28. The counts of eligible, responding and specific structure-space metrics by family and step, with the panel statistic and the four sensitivity sets: Table S29.
- *Specificity ratio.* The ratio, its interval and the verdict of every eligible metric at every step: Table S33 (comparators: Table S34); the 97 cells that survive the controls: Table S35; R per step on the 143 main enzymes with 90% intervals: Table S36; the robustness to motif size, packing matching and count gating: Table S31; the panel statistic against the earlier whole-panel analysis: Table S42; the effective number of independent metrics: Table S46.
- *Controls and floors.* The loosest matching tier needed, and the balance of residues changed in the two arms: Tables S26 and S27; the burial of the constructed control in the de novo designs: Table S47; the sanity floors: Table S25.
- *Predictor ladder.* Every metric at every step: Table S43; the three controls and the pre-specified rule: Table S30; the distance of each pair to the ligand: Table S37; the ratio in distance strata: Table S39; the caliper-matched subsets: Table S40; the regression adjustment: Table S41; the 54 pairs given a new distance-matched control: Table S38; the seed noise: Table S32.
- *Not made.* Experiments that could not be made or were retired: Table S48; PLACER on the isosteric step: Table S49.

3.4.2 Detecting second shell lesion

Experiment. One, two or four residues of the second shell around the catalytic residues are mutated and the structure is repacked (second-shell removal); the control does the same for the same number of residues around matched non-catalytic positions, and both arms are scored over the catalytic residues.

Dataset. The 143 M-CSA enzymes [32] pooled with the 52 pilot enzymes that are not in it, with a control arm for up to 193 of them; the number of enzymes in which a metric moves at all ranges from 0 to 182 per metric and dose. Structure-space metrics are scored on this lesion (Methods 2.5.2).

Ranking criteria. The 29 structure-space main-set metrics are ranked as in Table 10, with R at the three doses (1, 2 and 4 residues removed) and its mean over the doses; "Detected from" is the smallest number of residues removed at which the metric responds ("Never" if none), and rows within a group are ranked by raw R. A metric with no R in at least five proteins has "-". Raw values are reported (see the caption); the last column counts, for each metric, the doses (of 3) at which it responds to the removal and at which the control response is above the metric's noise.

Detected from	Rank in group	Item	Group	R at 1 / 2 / 4 residues removed	Mean R [95% interval]	Doses with a response / with a control above noise
4 residues	1	pKa shift, mean (PROPKA) [25, 26]	pKa	0.74 / 0.72 / 0.64	0.70 [0.64, 0.75]	1 / 0
4 residues	2	pKa, mean (PROPKA) [25, 26]	pKa	0.73 / 0.71 / 0.64	0.69 [0.63, 0.75]	1 / 0
Never	1	Fraction buried	Geometry	0.91 / 0.79 / 0.65	0.78 [0.68, 0.87]	0 / 0
Never	2	Rotamer, Dunbrack (Rosetta, site) [20]	Rosetta energy	0.79 / 0.76 / 0.69	0.75 [0.69, 0.79]	0 / 0
Never	3	pKa, max (PROPKA) [25, 26]	pKa	0.75 / 0.72 / 0.68	0.71 [0.66, 0.77]	0 / 0
Never	4	Pairwise distance, mean	Geometry	0.76 / 0.75 / 0.63	0.71 [0.65, 0.77]	0 / 0
Never	5	Radius of gyration	Geometry	0.77 / 0.71 / 0.64	0.71 [0.64, 0.76]	0 / 0
Never	6	pKa, range (PROPKA) [25, 26]	pKa	0.74 / 0.72 / 0.65	0.71 [0.65, 0.76]	0 / 0
Never	7	lk-ball solvation (Rosetta, site) [20]	Rosetta energy	0.72 / 0.73 / 0.66	0.70 [0.66, 0.75]	0 / 1
Never	8	Total SASA	Geometry	0.73 / 0.71 / 0.66	0.70 [0.65, 0.75]	0 / 0
Never	9	Intra-residue repulsion (Rosetta, site) [20]	Rosetta energy	0.72 / 0.73 / 0.64	0.70 [0.64, 0.75]	0 / 0
Never	10	pKa shift, max absolute (PROPKA) [25, 26]	pKa	0.75 / 0.68 / 0.66	0.70 [0.64, 0.75]	0 / 1
Never	11	Relative SASA, mean	Geometry	0.73 / 0.71 / 0.64	0.70 [0.64, 0.74]	0 / 0
Never	12	Attraction (Rosetta, site) [20]	Rosetta energy	0.73 / 0.69 / 0.61	0.67 [0.63, 0.72]	0 / 0
Never	13	Relative SASA, max	Geometry	0.69 / 0.69 / 0.64	0.67 [0.61, 0.73]	0 / 0
Never	14	Electrostatics (Rosetta, site) [20]	Rosetta energy	0.74 / 0.70 / 0.58	0.67 [0.62, 0.72]	0 / 1
Never	15	Worst residue (Rosetta, site) [20]	Rosetta energy	0.74 / 0.68 / 0.59	0.67 [0.61, 0.72]	0 / 0
Never	16	pKa, min (PROPKA) [25, 26]	pKa	0.74 / 0.67 / 0.60	0.67 [0.61, 0.73]	0 / 0
Never	17	Net van der Waals (Rosetta, site) [20]	Rosetta energy	0.74 / 0.67 / 0.58	0.66 [0.61, 0.71]	0 / 1
Never	18	H-bond, side chain (Rosetta, site) [20]	Rosetta energy	0.72 / 0.68 / 0.59	0.66 [0.59, 0.72]	0 / 1
Never	19	Site total (Rosetta) [20]	Rosetta energy	0.71 / 0.66 / 0.60	0.66 [0.60, 0.71]	0 / 1
Never	20	Pairwise distance, max	Geometry	0.72 / 0.65 / 0.60	0.66 [0.52, 0.76]	0 / 0
Never	21	Repulsion (Rosetta, site) [20]	Rosetta energy	0.71 / 0.63 / 0.62	0.65 [0.60, 0.71]	0 / 1
Never	22	Solvation (Rosetta, site) [20]	Rosetta energy	0.72 / 0.66 / 0.57	0.65 [0.60, 0.70]	0 / 1
Never	23	H-bond, backbone to side chain (Rosetta, site) [20]	Rosetta energy	0.56 / 0.68 / 0.65	0.63 [0.56, 0.70]	0 / 0
Never	24	Polar contacts	Geometry	0.59 / 0.59 / 0.65	0.61 [0.49, 0.73]	0 / 0
Never	25	Ramachandran (Rosetta, site) [20]	Rosetta energy	0.10 / 0.00 / 0.00	0.03 [0.00, 0.11]	0 / 3

Detected from	Rank in group	Item	Group	R at 1 / 2 / 4 residues removed	Mean R [95% interval]	Doses with a response / with a control above noise
Never	–	Omega torsion (Rosetta, site) [20]	Rosetta energy	– / – / –	–	0 / 3
Never	–	Amino-acid propensity (Rosetta, site) [20]	Rosetta energy	– / – / –	–	0 / 3

Table 11. Second-shell lesion: the 29 structure-space main-set metrics grouped by the dose at which they are first detected and ranked by raw R within the group. Rank statistic R, the probability that a metric changes more when second-shell residues are removed around the catalytic residues than around a matched control (0.5 = no specificity), at the three doses and as the mean over the doses, with a 95% interval that resamples enzyme sub-subclasses. "Detected from" is the smallest number of residues removed at which the metric responds ("Never" if none); a metric with no R in at least five proteins has "–". No cell qualifies as interpretable, so raw values are reported: the last column counts, for each metric, the doses (of 3) at which it responds to the removal and at which the control response is above the metric's noise. The 8 supplementary comparators and the per-dose flags are in Table S44. Source: main/T11_specificity_second_shell_lesion.csv.

More detail. The 37 metrics scored for the second-shell lesion, with the 8 supplementary comparators and the response and control-noise flags at each dose: Table S44. The specificity ratio and verdict at each dose: Table S33 (comparators: Table S34). The counts of eligible, responding and specific metrics by family and dose: Table S29.

4 Discussion

As generative models produce more designs than any laboratory can test, the in-silico filter that chooses among them has become a decisive step of enzyme design. The designs that are built are the ones that pass it, and the same metrics increasingly decide which design methods are judged better. A filter that does not track catalysis therefore wastes experiments and can mislead the field about what works, yet the metrics behind these filters had not, to our knowledge, been tested against catalysis.

We tested 33 such metrics in four tests, each corresponding to a job that a filter is asked to do. Activity discrimination is the basic one, to separate designs that work from designs that do not. Substrate discrimination asks whether a metric knows what an enzyme is built for, which design conditioned on a substrate takes for granted. Activity ranking is the last step of a selection, choosing the best of the designs that pass. Detection ability asks for the property that makes a metric a measure of catalysis and not of geometry: it must register damage to the catalytic residues more than the same damage elsewhere in the protein. Three of the tests observe proteins as they exist and the fourth intervenes, which is what allows catalysis to be told apart from proximity. In every test a metric has to beat a trivial reference before it counts as having detected anything. A metric that tracks catalysis should do well in all four.

In activity discrimination, pLDDT of the whole prediction ranked first on natural dead and active enzymes (AUROC 0.79) and the best of the 33 metrics, the intra-residue repulsion of the catalytic residues, reached 0.61, below the chance threshold of 0.68; on 192 designs the intra-residue repulsion ranked first (0.80) and the tyrosine fraction reached 0.75. In substrate discrimination, ipTM ranked first (0.70) and the RMSD of the catalytic atoms reached 0.53 to 0.56. In activity ranking, the range of the catalytic pKa correlated best with the measured impairment of BglB variants ($\rho = 0.48$) and the distance from the mutated residue to the catalytic residues gave 0.46; no metric beat it. In detection ability, the per-chain pTM minimum responded most to catalytic damage ($R = 0.75$) and gave 0.59 when the control lay at the same distance from the ligand. The 192 designs, of which 16 are active, had already passed the filters of the campaign that made them.

References

- [1] Listov, D., Fleishman, S. J. De novo enzyme design: Controlling structure to design function. *Curr. Opin. Struct. Biol.* 98, 103252 (2026). doi:10.1016/j.sbi.2026.103252
- [2] Butcher, J., Krishna, R., Mitra, R. et al. De novo Design of All-atom Biomolecular Interactions with RFdiffusion3. *bioRxiv* (2025). doi:10.1101/2025.09.18.676967
- [3] Kim, D., Woodbury, S. M., Ahern, W. et al. Computational design of metallohydrolases. *Nature* 649, 246–253 (2026). doi:10.1038/s41586-025-09746-w
- [4] Listov, D., Vos, E., Hoffka, G. et al. Complete computational design of high-efficiency Kemp elimination enzymes. *Nature* 643, 1421–1427 (2025). doi:10.1038/s41586-025-09136-2

- [5] Richter, F., Leaver-Fay, A., Khare, S. D., Bjelic, S., Baker, D. De novo enzyme design using Rosetta3. *PLOS ONE* 6, e19230 (2011). doi:10.1371/journal.pone.0019230
- [6] Abramson, J., Adler, J., Dunger, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* 630, 493–500 (2024). doi:10.1038/s41586-024-07487-w
- [7] Lauko, A., Pellock, S. J., Sumida, K. H. et al. Computational design of serine hydrolases. *Science* 388, eadu2454 (2025). doi:10.1126/science.adu2454
- [8] Ahern, W., Yim, J., Tischler, D. et al. Atom-level enzyme active site scaffolding using RFdiffusion2. *Nature Methods* 23, 96–105 (2026). doi:10.1038/s41592-025-02975-x
- [9] Watson, J. L., Juergens, D., Bennett, N. R. et al. De novo design of protein structure and function with RFdiffusion. *Nature* 620, 1089–1100 (2023). doi:10.1038/s41586-023-06415-8
- [10] Yeh, A. H., Norn, C., Kipnis, Y. et al. De novo design of luciferases using deep learning. *Nature* 614, 774–780 (2023). doi:10.1038/s41586-023-05696-3
- [11] Braun, M., Tripp, A., Chakotok, M. et al. Computational enzyme design by catalytic motif scaffolding. *Nature* 649, 237–245 (2026). doi:10.1038/s41586-025-09747-9
- [12] Jumper, J., Evans, R., Pritzel, A. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* 596, 583–589 (2021). doi:10.1038/s41586-021-03819-2
- [13] Chai Discovery, Boitreaud, J., Dent, J. et al. Chai-1: Decoding the molecular interactions of life. *bioRxiv* (2024). doi:10.1101/2024.10.10.615955
- [14] Anishchenko, I., Kipnis, Y., Kalvet, I. et al. Modeling protein-small molecule conformational ensembles with PLACER. *Proc. Natl. Acad. Sci. U.S.A.* 122, e2427161122 (2025). doi:10.1073/pnas.2427161122
- [15] Johnson, S. R., Fu, X., Viknander, S. et al. Computational scoring and experimental evaluation of enzymes generated by neural networks. *Nature Biotechnology* 43, 396–405 (2025). doi:10.1038/s41587-024-02214-2
- [16] Overath, M. D., Rygaard, A. S. H., Jacobsen, C. P. et al. Predicting Experimental Success in De Novo Binder Design: A Meta-Analysis of 3,766 Experimentally Characterised Binders. *bioRxiv* (2025). doi:10.1101/2025.08.14.670059
- [17] Garcia, M., Dixit, S. M., Rocklin, G. J. Evaluating zero-shot prediction of monomeric protein design success by AlphaFold, ESMFold, and ProteinMPNN. *Protein Sci.* 35, e70453 (2026). doi:10.1002/pro.70453
- [18] Korbeld, K. T., Viliuga, V., Fürst, M. J. L. J. Limitations of the refolding pipeline for de novo protein design. *Protein Sci.* 35, e70613 (2026). doi:10.1002/pro.70613
- [19] Kiss, G., Röthlisberger, D., Baker, D., Houk, K. N. Evaluation and ranking of enzyme designs. *Protein Sci.* 19, 1760–1773 (2010). doi:10.1002/pro.462
- [20] Alford, R. F., Leaver-Fay, A., Jeliazkov, J. R. et al. The Rosetta All-Atom Energy Function for Macromolecular Modeling and Design. *J. Chem. Theory Comput.* 13, 3031–3048 (2017). doi:10.1021/acs.jctc.7b00125
- [21] Chaudhury, S., Lyskov, S., Gray, J. J. PyRosetta: a script-based interface for implementing molecular modeling algorithms using Rosetta. *Bioinformatics* 26, 689–691 (2010). doi:10.1093/bioinformatics/btq007
- [22] Lee, B., Richards, F. M. The interpretation of protein structures: Estimation of static accessibility. *Journal of Molecular Biology* 55, 379–400 (1971). doi:10.1016/0022-2836(71)90324-X
- [23] Mitternacht, S. FreeSASA: An open source C library for solvent accessible surface area calculations. *F1000Research* 5, 189 (2016). doi:10.12688/f1000research.7931.1
- [24] Tien, M. Z., Meyer, A. G., Sydykova, D. K., Spielman, S. J., Wilke, C. O. Maximum Allowed Solvent Accessibilities of Residues in Proteins. *PLoS ONE* 8, e80635 (2013). doi:10.1371/journal.pone.0080635
- [25] Olsson, M. H. M., Søndergaard, C. R., Rostkowski, M., Jensen, J. H. PROPKA3: Consistent Treatment of Internal and Surface Residues in Empirical pKa Predictions. *Journal of Chemical Theory and Computation* 7, 525–537 (2011). doi:10.1021/ct100578z
- [26] Søndergaard, C. R., Olsson, M. H. M., Rostkowski, M., Jensen, J. H. Improved Treatment of Ligands and Coupling Effects in Empirical Calculation and Rationalization of pKa Values. *Journal of Chemical Theory and Computation* 7, 2284–2295 (2011). doi:10.1021/ct200133y
- [27] Evans, R., O'Neill, M., Pritzel, A. et al. Protein complex prediction with AlphaFold-Multimer. *bioRxiv* (2021). doi:10.1101/2021.10.04.463034
- [28] Burley, S. K., Bhatt, R., Bhikadiya, C. et al. Updated resources for exploring experimentally-determined PDB structures and Computed Structure Models at the RCSB Protein Data Bank. *Nucleic Acids Research* 53, D564–D574 (2025). doi:10.1093/nar/gkae1091
- [29] Berman, H. M., Westbrook, J., Feng, Z. et al. The Protein Data Bank. *Nucleic Acids Research* 28, 235–242 (2000). doi:10.1093/nar/28.1.235
- [30] The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research* 51, D523–D531 (2023). doi:10.1093/nar/gkac1052
- [31] Dana, J. M., Gutmanas, A., Tyagi, N. et al. SIFTS: updated Structure Integration with Function, Taxonomy and Sequences resource allows 40-fold increase in coverage of structure-based annotations for proteins. *Nucleic Acids Research* 47, D482–D489 (2019). doi:10.1093/nar/gky1114
- [32] Ribeiro, A. J. M., Holliday, G. L., Furnham, N., Tyzack, J. D., Ferris, K., Thornton, J. M. Mechanism and Catalytic Site Atlas (M-CSA): a database of enzyme reaction mechanisms and active sites. *Nucleic Acids Research* 46, D618–D623 (2018). doi:10.1093/nar/gkx1012

- [33] Hanley, J. A., McNeil, B. J. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology* 143, 29–36 (1982). doi:10.1148/radiology.143.1.7063747
- [34] Mann, H. B., Whitney, D. R. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics* 18, 50–60 (1947). doi:10.1214/aoms/1177730491
- [35] Efron, B. Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics* 7, 1–26 (1979). doi:10.1214/aos/1176344552
- [36] Field, C. A., Welsh, A. H. Bootstrapping Clustered Data. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 69, 369–390 (2007). doi:10.1111/j.1467-9868.2007.00593.x
- [37] Westfall, P. H., Young, S. S. *Resampling-Based Multiple Testing: Examples and Methods for p-Value Adjustment*. Wiley (1993).
- [38] Carlin, D. A., Caster, R. W., Wang, X. et al. Kinetic Characterization of 100 Glycoside Hydrolase Mutants Enables the Discovery of Structural Features Correlated with Kinetic Constants. *PLOS ONE* 11, e0147596 (2016). doi:10.1371/journal.pone.0147596
- [39] Carlin, D. A., Hapig-Ward, S., Chan, B. W. et al. Thermal stability and kinetic constants for 129 variants of a family 1 glycoside hydrolase reveal that enzyme activity and stability can be separately designed. *PLOS ONE* 12, e0176255 (2017). doi:10.1371/journal.pone.0176255
- [40] Henikoff, S., Henikoff, J. G. Amino acid substitution matrices from protein blocks. *Proceedings of the National Academy of Sciences* 89, 10915–10919 (1992). doi:10.1073/pnas.89.22.10915
- [41] Spearman, C. The proof and measurement of association between two things. *The American Journal of Psychology* 15, 72–101 (1904). doi:10.2307/1412159